Vietnam Math Modeling Competition 2026

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SECOND ROUND – PART A

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ORGANIZER

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Acknowledgments

Our team would like to express our sincere gratitude to the Organizing Committee and the Board of Judges for their dedicated work in building a serious academic competition, where students can approach applied mathematics not as dry formulas, but as a living tool for understanding and solving a real problem. The problem statement itself posed an interesting challenge for us: how to turn assumptions about tires, pit time and stint length into a quantitative model that is rigorous enough to be trusted, yet flexible enough to reflect the uncertainty of a real race.

Throughout the process, we learned more than an ordinary exam could offer. The team practiced turning data and assumptions into a mathematical model and, most importantly, learned how to connect a mathematical result to a meaningful strategic decision. The debates to agree on a single number and the nights spent revising the model after discovering a gap in the reasoning were the most memorable parts of this journey.

Introduction

In a Formula 1 race, victory often belongs not to the fastest car, but to the team that makes the best tire decisions. This report considers the problem of choosing the tire compound, the length of each stint and the number of pit stops for a race of (N) laps. The three compounds (S,M,H) create a trade-off problem. A soft tire may give a high base speed but wears quickly and reaches the performance cliff early, while a hard tire is more durable but slower. Every pit stop to change tires costs a fixed amount of time, so the optimal strategy is a delicate balance between speed, durability and the number of stops.The three compounds S, M, H trade off base speed, wear rate and the performance cliff. The model starts from the total stint time $S_c(n)$ and extends to multi-compound optimization, pit-threshold analysis, the F1 constraint, parameter uncertainty and calibration with real data.

Notation and Conventions

Symbol	Unit	Meaning
N	laps	Total number of laps.
P	s	Pit time.
c	–	Tire compound, $c \in \{S, M, H\}$.
S, M, H	–	The three compounds: Soft, Medium, Hard.
B_c	s	Base lap time of a new tire.
d_c	s/lap	Degradation before the cliff.
L_c	laps	Performance cliff.
D_c	s/lap	Degradation after the cliff.
$t_c(k)$	s	Lap k of a stint on tire c.
$S_c(n)$	s	Total of a stint of length n.
(c_i, n_i)	–	Stint i: tire and length.
m	stints	Number of stints; pits are $m-1$.
n_i	laps	Stint length, $\sum_i n_i = N$.
T	s	Total strategy time.
N_{pit}	count	Number of pit stops.
T_{base}	s	Physical baseline forecast.
T_{actual}	s	Observed time.
T_{pred}	s	Forecast after residual.
$\hat{r}$	s	Residual learned from context.
ε	s	Error: $T_{\text{actual}} - T_{\text{pred}}$.

Symbol	Unit	Meaning
z_j	–	Standardized variable $(x_j - \mu_j)/\sigma_j$.
$\bar{t}_c(n)$	s/lap	Stint average with pit amortized.
n_c^*	laps	Continuous optimal length.
m_c^*	stints	Corresponding continuous stint count.
$g(n)$	s	Best stint of length n.
$\delta(n)$	s	Marginal: $g(n) - g(n-1)$.
A_m	s	Optimum with exactly m stints, before pits.
$T_m(P)$	s	$A_m + (m-1)P$.
$C_c(n)$	s	Stint cost $S_c(n) + P$.
$a_c(n)$	s/lap	Average $C_c(n)/n$.
λ	s/lap	Minimum value of $a_c(n)$.
$p_c(n)$	s	Penalty $S_c(n) + P - n\lambda$.
R	s	Total penalty $\sum_i p_{c_i}(n_i)$.
Δ	s	Time change when moving a lap.
$\Delta_c(r)$	s	$\lambda - t_c(r)$.
$\text{VaR}_\alpha[T]$	s	Quantile at level α.
$\text{CVaR}_\alpha[T]$	s	Mean of the bad tail.
n_{mc}	count	Number of Monte Carlo samples.

Executive Summary

This report builds a quantitative model for tire strategy in a Formula 1 race. Each stint is described by the lap time $t_c(k)$, the stint total $S_c(n)$, the pit cost P, and the total-lap constraint $\sum_i n_i = N$. The problem has two layers: choose the sequence (c_i, n_i) to minimize the total time, then test the robustness of the strategy when the tire and pit parameters are uncertain.

Key Results

Part	Conclusion
A1	Closed form for $S_c(n)$ and the optimal stint length $n_c^* = \sqrt{2P/d_c}$. With $N = 50$, $P = 25$, the best single-compound strategy is **Soft 3 pits (13–13–12–12)**, $T \approx$ **4132.6** s.
A2	When multiple compounds are allowed, the optimal deterministic strategy is **S15** + **S15** + **M20**, $T =$ **4131** s, using the 15-lap Soft stint as the backbone.
A3	The optimal number of stints decreases stepwise as P increases; the decision threshold is $\Delta_m = A_m - A_{m+1}$. At $P = 25$ ($23.4 < P < 29$), the solution is **S15** + **S15** + **M20**, $T =$ **4131** s.
A4	Reduce the problem to minimizing the total penalty relative to the ideal stint, then solve greedily; this gives the optimal valid strategy on **all 20 tracks** (e.g. Bahrain **M16**+**M16**+**M15**+**S10**, $T =$ **5592.4** s).
A5	**S–M–M** minimizes both $\sigma =$ **40.0** s and $\text{CVaR}_{90} =$ **4233.8** s.
A6	Calibration from real data (Verstappen, 3 races in 2023): keep the physical baseline as the core and learn the residual from context, avoiding forcing every deviation into the tire parameters.

The strategy recommendation depends on the criterion. **All-Soft 4 stints** gives the smallest expectation (**4142.7** s) in the A5 simulation, but **S–M–M** ($S15 \cdot M18 \cdot M17$) has the lowest σ and CVaR_{90}, and is only about **4.6** s (0.11%) slower in expectation. Therefore, if the team prioritizes low tail risk and stable results, **S–M–M** is the more robust choice.

Executive Message

The fastest strategy on paper is not necessarily the best strategy under uncertainty. The model shows that stints should not be pushed right up to the cliff, because the pit-time savings can be wiped out by tire degradation and tail risk.

1 Some Fundamental Theorems

1.1 Task Map and Notation Conventions

The Part A problem statement is divided in this report as follows.

Item	Content addressed in the report
A1(a)	Derive the formula for the time of one stint $S_c(n)$, including the two cases before and after the cliff.
A1(b)	Optimize the number of stints/pit stops when only one compound is used.
A2	Optimize the strategy when multiple compounds may be used, via marginal cost and structured exhaustive search.
A3	Examine how the optimal strategy changes with the pit time P.
A4	Add the F1 rule requiring at least two tire compounds; rank strategies for many tracks.
A5(a)	Model the uncertainty of B_c, d_c, L_c, D_c and the pit time.
A5(b)	Choose a robust strategy by expectation, variance, VaR and CVaR.
A6(a)	Choose the races, car/driver and the procedure for extracting real data.
A6(b)	Estimate B_c, d_c, L_c, D_c after separating out fuel and track effects.
A6(c)	Evaluate the current model, state its strengths/weaknesses and add the baseline–luck–error model.

From here on, T is the total deterministic time of a strategy in A1–A5. When moving to real data in A6(c), the report uses T_{base} for the baseline forecast and does not mix it with T_{pred}. Specifically,

$$T_{\text{pred}} = T_{\text{base}} + \widehat{r}, \qquad \varepsilon = T_{\text{actual}} - T_{\text{pred}}.$$

Thus the same physical formula is not renamed between parts: T_{base} is exactly the "baseline" part of the model, while $\widehat{r}$ is the part additionally learned from the race context.

1.2 Stint Formulas and Basic Optimization

Definition 1.2.1 (Lap time and stint time)**.** For tire c, the tire age at lap k is k. The piecewise linear model used in A1–A5 is

$$t_c(k) = \begin{cases} B_c + d_c(k-1), & k \le L_c, \\ B_c + d_c(L_c-1) + D_c(k-L_c), & k > L_c. \end{cases}$$

The total time of a stint of length n laps is

$$S_c(n) = \sum_{k=1}^{n} t_c(k).$$

For a strategy $(c_i, n_i)_{i=1}^{m}$, the total deterministic time is

$$T = \sum_{i=1}^{m} S_{c_i}(n_i) + (m-1)P, \qquad \sum_{i=1}^{m} n_i = N.$$

Proposition 1.2.2 (Closed form of $S_c(n)$)**.** If $n \le L_c$ then

$$S_c(n) = nB_c + \frac{d_c}{2} n(n-1).$$

If $n > L_c$ then

$$S_c(n) = nB_c + \frac{d_c}{2}(L_c-1)(2n-L_c) + \frac{D_c}{2}(n-L_c)(n-L_c+1).$$

Proof. When $n \le L_c$,

$$S_c(n) = \sum_{k=1}^{n} \big(B_c + d_c(k-1)\big) = nB_c + d_c \sum_{k=1}^{n} (k-1) = nB_c + \frac{d_c}{2} n(n-1).$$

When $n > L_c$, split the sum into two parts:

$$\sum_{k=1}^{L_c} \big(B_c + d_c(k-1)\big) + \sum_{k=L_c+1}^{n} \big(B_c + d_c(L_c-1) + D_c(k-L_c)\big).$$

The first part equals $L_c B_c + \frac{d_c}{2} L_c(L_c-1)$. The second part equals

$$(n-L_c)\big(B_c + d_c(L_c-1)\big) + D_c \sum_{u=1}^{n-L_c} u.$$

Adding the two parts and using $\sum_{u=1}^{h} u = h(h+1)/2$ with $h = n - L_c$ gives the stated formula. □

Lemma 1.2.3 (Marginal cost balance)**.** Consider a strategy with a fixed number of stints and a fixed compound for each stint. If there exist two stints i, j such that

$$t_{c_i}(n_i) > t_{c_j}(n_j + 1),$$

then moving one lap from stint i to stint j reduces the total tire running time. Therefore an interior optimal solution must satisfy

$$\max_i t_{c_i}(n_i) \le \min_j t_{c_j}(n_j + 1).$$

Proof. Keeping the number of stints and the compounds fixed, the pit part is constant. After moving one lap from i to j, the total tire time changes by

$$\big(S_{c_i}(n_i - 1) + S_{c_j}(n_j + 1)\big) - \big(S_{c_i}(n_i) + S_{c_j}(n_j)\big) = -t_{c_i}(n_i) + t_{c_j}(n_j + 1).$$

If $t_{c_i}(n_i) > t_{c_j}(n_j + 1)$ then this quantity is negative, contradicting optimality. The marginal balance condition is a direct consequence when no lap move can improve the solution. □

Proposition 1.2.4 (Continuous optimal stint length before the cliff)**.** When only one compound c is used and the stint lies before the cliff, the average cost per lap with the pit amortized is

$$\bar{t}_c(n) = \frac{S_c(n) + P}{n} = B_c + \frac{d_c}{2}(n-1) + \frac{P}{n}.$$

The continuous solution of the problem is

$$n_c^\star = \sqrt{\frac{2P}{d_c}}, \qquad m_c^\star = \frac{N}{n_c^\star} = N\sqrt{\frac{d_c}{2P}}.$$

If $n_c^\star > L_c$, the cliff constraint locks the solution to the feasible region near L_c; the integer solution is obtained by trying neighboring stint counts and balancing the lengths according to Lemma 1.2.3.

Proof. Take the derivative with respect to $n > 0$:

$$\frac{d}{dn}\bar{t}_c(n) = \frac{d_c}{2} - \frac{P}{n^2}.$$

Setting the derivative equal to 0 gives $n^2 = 2P/d_c$. The second derivative is $2P/n^3 > 0$, so this is a continuous minimum. Since the race has exactly N laps and the average stint length is $n_c^\star$, the continuous number of stints is $m_c^\star = N/n_c^\star$. The case beyond the cliff no longer satisfies the before-the-cliff assumption, so we must consider the boundary L_c and the nearest integer solutions. □

1.3 Probabilistic Foundations for A5

Definition 1.3.1 (Risk criteria)**.** For a random total time T, the risk quantile at level α is

$$\text{VaR}_\alpha[T] = \inf\{x : \Pr(T \le x) \ge \alpha\}.$$

The corresponding risk tail is

$$\text{CVaR}_\alpha[T] = \mathbb{E}[T \,|\, T \ge \text{VaR}_\alpha[T]],$$

or, in Monte Carlo simulation, the mean of the samples in the worst $1-\alpha$ group.

1.4 Probability Distributions and Foundational Lemmas

The notation $X \sim \mathscr{D}(\cdot)$ is read as "X follows the distribution $\mathscr{D}$ with the parameters written in parentheses". For example, $X \sim \text{Norm}(\mu, \sigma^2)$ means that X is a normally distributed random variable with expectation μ and variance σ^2. Below are the distributions used in the report, each with its formula and remarks.

(a) Uniform — $\text{Unif}(0,1)$**.** Density $f(u) = 1$ for $u \in [0,1]$ (every value in $[0,1]$ is equally likely). Remark: this is the basic random "seed"; from one $U \sim \text{Unif}(0,1)$ we can generate every other distribution through the quantile function (see item (d)).

(b) Normal — $\text{Norm}(\mu, \sigma^2)$**.** Density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}, \qquad x \in \mathbb{R}.$$

Remark: bell-shaped, symmetric about μ; expectation μ, variance σ^2; takes all real values (unbounded). $N(0,1)$ is the standard normal (μ=0, σ=1).

(b′) A shared tool: the moment generating function of the normal distribution.. To prove (not just state) the expectation/variance of the log-normal in item (c), and to add normal noises in §6.4, we prove one result and then reuse it several times. The moment generating function of a variable Y is the function $M_Y(t) := \mathbb{E}[e^{tY}]$ — a "barcode" of the distribution: two variables with the same MGF have the same distribution (we will use this property).

Proposition 1.4.1 (Moment generating function of the normal)**.** If $Y \sim \text{Norm}(\mu, \sigma^2)$ then for every real number t,

$$M_Y(t) = \mathbb{E}[e^{tY}] = \exp\Big(\mu t + \tfrac{1}{2}\sigma^2 t^2\Big).$$

Proof. By the definition of the expectation of a function of Y (integrating the value e^{ty} times the density),

$$\begin{aligned}\mathbb{E}[e^{tY}] &= \int_{-\infty}^{\infty} e^{ty}\,\frac{1}{\sigma\sqrt{2\pi}}\,e^{-(y-\mu)^2/(2\sigma^2)}\,dy \\ &= \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}}\,e^{ty-(y-\mu)^2/(2\sigma^2)}\,dy.\end{aligned}$$

We treat the exponent $S := ty - \dfrac{(y-\mu)^2}{2\sigma^2}$ separately using the technique of completing the square (high-school level): add and subtract a constant to collect y into a square. Specifically, we look for a constant a such that $S = -\dfrac{(y-a)^2}{2\sigma^2} + C$. Expanding both sides and matching the coefficients of y:

$$-\frac{(y-a)^2}{2\sigma^2} = -\frac{y^2}{2\sigma^2} + \frac{a}{\sigma^2}\,y - \frac{a^2}{2\sigma^2}, \qquad S = -\frac{y^2}{2\sigma^2} + \Big(\frac{\mu}{\sigma^2} + t\Big)y - \frac{\mu^2}{2\sigma^2}.$$

The coefficients of y on both sides are equal $\Rightarrow \dfrac{a}{\sigma^2} = \dfrac{\mu}{\sigma^2} + t \Rightarrow a = \mu + \sigma^2 t$. The remaining constant part:

$$C = \frac{a^2-\mu^2}{2\sigma^2} = \frac{(\mu+\sigma^2 t)^2 - \mu^2}{2\sigma^2} = \frac{2\mu\sigma^2 t + \sigma^4 t^2}{2\sigma^2} = \mu t + \tfrac{1}{2}\sigma^2 t^2.$$

Hence $S = -\dfrac{(y-a)^2}{2\sigma^2} + \big(\mu t + \frac{1}{2}\sigma^2 t^2\big)$, and taking the factor that does not depend on y outside the integral:

$$\mathbb{E}[e^{tY}] = e^{\mu t + \frac{1}{2}\sigma^2 t^2} \underbrace{\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}}\,e^{-(y-a)^2/(2\sigma^2)}\,dy}_{=1 \text{ (the area under a normal density always equals 1)}} = e^{\mu t + \frac{1}{2}\sigma^2 t^2}.$$

□

Proposition 1.4.2 (Expectation and variance of the log-normal)**.** If $X \sim \text{LogN}(\mu, \sigma^2)$ (i.e. $X = e^Y$ with $Y \sim \text{Norm}(\mu, \sigma^2)$) then

$$\mathbb{E}[X] = e^{\mu + \sigma^2/2}, \qquad \text{Var}[X] = \big(e^{\sigma^2} - 1\big)\,e^{2\mu+\sigma^2}.$$

Proof. Key point: $X = e^Y$ and $X^2 = e^{2Y}$, so their expectations are exactly the MGF of Y at $t = 1$ and $t = 2$. From Proposition 1.4.1:

$$\mathbb{E}[X] = \mathbb{E}[e^Y] = M_Y(1) = e^{\mu + \frac{1}{2}\sigma^2}, \qquad \mathbb{E}[X^2] = \mathbb{E}[e^{2Y}] = M_Y(2) = e^{2\mu + 2\sigma^2}.$$

Using the formula $\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$:

$$\text{Var}[X] = e^{2\mu+2\sigma^2} - \big(e^{\mu+\frac{1}{2}\sigma^2}\big)^2 = e^{2\mu+2\sigma^2} - e^{2\mu+\sigma^2} = e^{2\mu+\sigma^2}\big(e^{\sigma^2} - 1\big).$$

□

Proposition 1.4.3 (Linear combination of independent normals)**.** If $\xi \sim \text{Norm}(0,1)$ and $\nu \sim \text{Norm}(0,1)$ are independent, then for all constants a, b,

$$W := a\xi + b\nu \;\sim\; \text{Norm}\big(0,\, a^2 + b^2\big); \qquad \mathbb{E}[W] = 0, \quad \text{Var}[W] = a^2 + b^2.$$

Proof. Variance (the part we actually use to calibrate the coefficients): for two independent variables, use the rule $\text{Var}[aX + bY] = a^2\text{Var}[X] + b^2\text{Var}[Y] + 2ab\,\text{Cov}(X,Y)$ and $\text{Cov}(\xi, \nu) = 0$ (by independence):

$$\text{Var}[W] = a^2 \cdot 1 + b^2 \cdot 1 + 0 = a^2 + b^2.$$

Normal form: use Proposition 1.4.1 (each variable has $\mu{=}0, \sigma{=}1$) and independence (the expectation of a product equals the product of the expectations):

$$M_W(t) = \mathbb{E}\big[e^{t(a\xi+b\nu)}\big] = \mathbb{E}[e^{ta\xi}]\,\mathbb{E}[e^{tb\nu}] = e^{\frac{1}{2}a^2t^2}\,e^{\frac{1}{2}b^2t^2} = e^{\frac{1}{2}(a^2+b^2)t^2},$$

which is exactly the MGF of $\text{Norm}(0, a^2+b^2)$. Since the MGF uniquely determines the distribution, W is normal with variance $a^2 + b^2$. □

(c) Log-normal — LogN(μ, σ^2).. Definition: $X \sim \text{LogN}(\mu, \sigma^2)$ if and only if $\ln X \sim \text{Norm}(\mu, \sigma^2)$. Why it is called this and why it is useful: the condition "$\ln X$ is normal" is equivalent to $X = e^Y$ with $Y \sim \text{Norm}(\mu, \sigma^2)$ — i.e. X is e raised to the power of a normal variable. Two properties follow immediately:

- Always positive: $e^Y > 0$ for every $Y \in \mathbb{R}$, so $X > 0$ — suitable for quantities that cannot be negative.
- Right-skewed: the function $e^{(\cdot)}$ stretches the upper half (the $Y > 0$) and compresses the lower half (the $Y < 0$), creating a long right tail.

Density (change of variable $x = e^y \Rightarrow y = \ln x$, $|dy/dx| = 1/x$): multiply the normal density of Y by $|dy/dx|$,

$$\begin{aligned}f_X(x) &= \frac{1}{\sigma\sqrt{2\pi}}e^{-(\ln x-\mu)^2/(2\sigma^2)} \cdot \frac{1}{x} \\ &= \frac{1}{x\,\sigma\sqrt{2\pi}}\,e^{-(\ln x-\mu)^2/(2\sigma^2)}, \qquad x > 0.\end{aligned}$$

Expectation and variance (proved in Proposition 1.4.2): $\mathbb{E}[X] = e^{\mu+\sigma^2/2}$, $\text{Var}[X] = (e^{\sigma^2}-1)\,e^{2\mu+\sigma^2}$.

Coefficient of variation (CoV).. For any random variable X,

$$\text{CoV}(X) := \frac{\sigma_X}{\mu_X} = \frac{\text{standard deviation}}{\text{expectation}}.$$

The CoV expresses relative dispersion — the typical error relative to the mean value itself, in dimensionless form (a ratio, usually written as a percentage). For example, CoV $= 0.5$ means that the standard deviation equals 50% of the expectation. Advantage: it allows comparing the uncertainty levels of quantities with different units (seconds versus laps). Applying the two expectation–variance formulas above to the log-normal:

$$\text{CoV} = \frac{\sqrt{(e^{\sigma^2}-1)\,e^{2\mu+\sigma^2}}}{e^{\mu+\sigma^2/2}} = \sqrt{e^{\sigma^2}-1} \approx \sigma \quad (\text{when } \sigma \text{ is small}).$$

That is, the parameter σ (the deviation on the log scale) is approximately equal to the CoV itself — very convenient for setting a desired level of relative uncertainty.

(d) Logistic and Log-logistic.. The logistic distribution resembles the normal (symmetric, bell-shaped) but has heavier tails, and is defined by the cumulative distribution function $F(x) = 1/\big(1 + e^{-(x-\mu)/s}\big)$. The log-logistic distribution is the "logistic on the log scale" (just as the log-normal arises from the normal): $X > 0$ with the cumulative distribution function

$$F(x) = \frac{(x/a)^\beta}{1 + (x/a)^\beta}, \qquad x > 0,$$

where $a > 0$ is the scale parameter (setting the location of the distribution) and $\beta > 0$ is the shape parameter — this is exactly the β that appears in the sampling formula. Meaning of β: large $\beta \Rightarrow$ light tail, concentrated distribution; small $\beta \Rightarrow$ heavy right tail (extremely large values are more likely). Remark: positive, right-skewed, polynomial tail.

Deriving the sampling formula (quantile function = inverse of F). A basic property: if $U \sim \text{Unif}(0,1)$ then $X = F^{-1}(U)$ has exactly the distribution with cumulative function F (the inverse transform method). We solve $U = F(X)$:

$$\begin{aligned}U = \frac{(X/a)^\beta}{1+(X/a)^\beta} \;&\Rightarrow\; U\big(1+(X/a)^\beta\big) = (X/a)^\beta, \\ (X/a)^\beta = \frac{U}{1-U} \;&\Rightarrow\; X = a\Big(\frac{U}{1-U}\Big)^{1/\beta}.\end{aligned}$$

Adding a floor P_0 (the physical minimum value) and renaming the scale $a \to m$, we obtain exactly the formula used in (7): $P = P_0 + m\,(U/(1-U))^{1/\beta}$. At $U = 0.5$ we have $U/(1-U) = 1$ so $X = a$, i.e. the median equals $P_0 + m$.

1.5 Expectation and Variance of B_c

Let $W := a_{\text{ht}}^B \xi + a_{\text{rg}}^B \nu_c$. By Proposition 1.4.3, $W \sim \text{Norm}\big(0, (a_{\text{ht}}^B)^2 + (a_{\text{rg}}^B)^2\big)$, so $\mathbb{E}[W] = 0$ and $\text{Var}[W] = (a_{\text{ht}}^B)^2 + (a_{\text{rg}}^B)^2$. Since $B_c = B_c^0(1+W)$ and B_c^0 is a constant:

$$\begin{aligned}\mathbb{E}[B_c] &= B_c^0\big(1+\mathbb{E}[W]\big) = B_c^0,\\ \text{Var}[B_c] &= (B_c^0)^2\,\text{Var}[W] = (B_c^0)^2\big((a_{\text{ht}}^B)^2 + (a_{\text{rg}}^B)^2\big),\end{aligned}$$

where $\text{Var}[cZ] = c^2\text{Var}[Z]$ with the constant $c = B_c^0$. Hence the standard deviation and CoV:

$$\sigma_{B_c} = B_c^0\sqrt{(a_{\text{ht}}^B)^2 + (a_{\text{rg}}^B)^2}, \qquad \text{CoV}(B_c) = \frac{\sigma_{B_c}}{\mathbb{E}[B_c]} = \sqrt{(a_{\text{ht}}^B)^2 + (a_{\text{rg}}^B)^2}.$$

Two notable consequences: (1) the expectation equals exactly the nominal value B_c^0 — the noise is unbiased; (2) the CoV does not depend on the compound, so the absolute deviation $\sigma_{B_c} = B_c^0 \cdot \text{CoV}$ is proportional to the pace — exactly the nature of "scaling proportionally".

1.6 Delta-Method Confidence Band

Theorem 1.6.1 (Variance-propagation confidence band)**.** Suppose B_c, d_c, D_c, L_c are independent with standard deviations $\sigma_B, \sigma_d, \sigma_D, \sigma_L$. Then, to first order (delta-method), the variance of the lap time at tire age k is

$$\begin{aligned}\text{Var}[t_c(k)] \approx \sigma_B^2 + (k-1)^2\sigma_d^2 + \mathbf{1}_{k>L_c}(k-L_c)^2\sigma_D^2 \\ + \mathbf{1}_{|k-L_c|\le 1}(D_c-d_c)^2\sigma_L^2,\end{aligned} \tag{1}$$

and the 95% confidence band is $t_c(k) \pm 1.96\sqrt{\text{Var}[t_c(k)]}$.

Proof. Treat (2) as a function of the random variables; the partial derivatives at the nominal values are: $\partial/\partial B = 1$, $\partial/\partial d = (k-1)$ (before the cliff), $\partial/\partial D = (k-L_c)^+$. At the cliff the function has a kink with respect to L_c; the delta-method assigns the contribution $(D_c-d_c)^2\sigma_L^2$ within a one-lap neighborhood of L_c. Adding the terms $\big(\partial f/\partial\theta\big)^2\sigma_\theta^2$ (independent) gives (1). □

1.7 Safety-Margin Theorem Around the Cliff

Theorem 1.7.1 (Asymmetry around the cliff ⇒ safety margin)**.** Consider a stint of length n on a tire with a random cliff L. The expected time of the last lap is

$$\mathbb{E}[t_c(n)] = B_c + d_c(n-1) + (D_c-d_c)\,\varphi(n), \qquad \varphi(n) := \mathbb{E}[(n-L)^+].$$

The penalty function φ is convex and nondecreasing, with difference $\varphi(n) - \varphi(n-1) = F_L(n-1)$ (the cumulative distribution function of L). Since $D_c - d_c > 0$ and F_L rises from 0 to 1 around the median of L, the expected marginal cost accelerates as n approaches the nominal cliff. Consequence: the robust optimal stint length is strictly smaller than the deterministic length $n^\star$ — a safety margin.

Proof. Let $p_L(\ell) := \Pr(L = \ell)$ be the probability mass function of the discrete variable L — i.e. the probability that the cliff falls exactly on lap ℓ — with $\sum_\ell p_L(\ell) = 1$. By the definition of the expectation of a function of L,

$$\varphi(n) = \mathbb{E}[(n-L)^+] = \sum_\ell p_L(\ell)\,(n-\ell)^+.$$

Take the first-order difference and bring the sum outside:

$$\varphi(n) - \varphi(n-1) = \sum_\ell p_L(\ell)\Big[(n-\ell)^+ - (n-1-\ell)^+\Big].$$

Consider the bracketed quantity $g_\ell := (n-\ell)^+ - (n-1-\ell)^+$ according to the position of ℓ:

- **If $\ell \le n-1$:** then $n-\ell \ge 1 > 0$ so $(n-\ell)^+ = n-\ell$; and $n-1-\ell \ge 0$ so $(n-1-\ell)^+ = n-1-\ell$. Hence $g_\ell = (n-\ell) - (n-1-\ell) = 1$.
- **If $\ell \ge n$:** then $n-\ell \le 0$ and $n-1-\ell \le -1 < 0$, so both positive parts equal 0: $g_\ell = 0 - 0 = 0$.

In summary $g_\ell = \mathbf{1}_{\{\ell \le n-1\}}$ (equal to 1 if $\ell \le n-1$, and 0 otherwise). Substituting:

$$\begin{aligned}\varphi(n) - \varphi(n-1) &= \sum_\ell p_L(\ell)\,\mathbf{1}_{\{\ell\le n-1\}}\\ &= \sum_{\ell\le n-1} p_L(\ell) = \Pr(L \le n-1) = F_L(n-1),\end{aligned}$$

where F_L is the cumulative distribution function of L. Since F_L is nondecreasing in its argument, the first-order difference $\varphi(n) - \varphi(n-1)$ increases with n — meaning that φ is discretely convex; and since $F_L \ge 0$, φ is nondecreasing.

Now for the meaning of the asymmetry. The expected marginal cost of the n-th lap is $\mathbb{E}[t_c(n)] - \mathbb{E}[t_c(n-1)] = d_c + (D_c - d_c)F_L(n-1)$. When n is still much smaller than the typical cliff, $F_L(n-1) \approx 0$ so the marginal cost is $\approx d_c$. When n passes the concentration region of L, $F_L(n-1) \to 1$ so the marginal cost rises to $\approx D_c$. Since $D_c \gg d_c$, ending the stint exactly at the nominal cliff carries a large expected penalty (in the scenarios where L turns out to be smaller), while backing off by one lap only saves $\approx d_c$. The imbalance between "a large penalty for overshooting" and "a small gain for caution" pushes the minimizer of $\mathbb{E}[t_c(n)]$ (and therefore of the expected total time) ahead of the deterministic length $n^\star$, creating a safety margin. □

1.8 Evaluation Foundations for A6(c)

Definition 1.8.1 (Baseline–luck–error)**.** In A6(c), the baseline model is written uniformly as

$$T_{\text{base}} = \sum_{\text{laps}} t_c(k) + N_{\text{pit}}P + T_{\text{fuel/track}} + T_{\text{first lap}}.$$

The old model is $T_{\text{pred}}^{(0)} = T_{\text{base}}$. The new model learns the residual

$$r_i = T_{\text{actual},i} - T_{\text{base},i}, \qquad T_{\text{pred},i}^{(1)} = T_{\text{base},i} + \widehat{r}_i.$$

Proposition 1.8.2 (Metrics for comparing the old and new models)**.** With the error $e_i^{(q)} = T_{\text{actual},i} - T_{\text{pred},i}^{(q)}$, $q \in \{0,1\}$, the main metrics are

$$\text{MAE}_q = \frac{1}{n}\sum_i |e_i^{(q)}|, \qquad \text{RMSE}_q = \sqrt{\frac{1}{n}\sum_i (e_i^{(q)})^2}, \qquad \bar{e}_q = \frac{1}{n}\sum_i e_i^{(q)}.$$

The improvement is read through

$$\begin{aligned}\Delta\text{MAE} &= \text{MAE}_0 - \text{MAE}_1,\\ \Delta\text{RMSE} &= \text{RMSE}_0 - \text{RMSE}_1, \qquad \Delta R^2 = R_1^2 - R_0^2.\end{aligned}$$

Proof. These formulas are the standard definitions of the mean absolute error, the root mean squared error and the mean bias. Since the new model is accepted only when the test error decreases, positive values of ΔMAE, ΔRMSE and ΔR^2 measure exactly the improvement beyond the baseline. □

Proposition 1.8.3 (Sensitivity of the baseline and the residual)**.** If $n \le L_c$, then

$$\frac{\partial S_c(n)}{\partial B_c} = n, \qquad \frac{\partial S_c(n)}{\partial d_c} = \frac{n(n-1)}{2}, \qquad \frac{\partial T}{\partial P} = N_{\text{pit}}.$$

If an exponential cliff is used and $h = n - L_c > 0$, then

$$\frac{\partial S_c(n)}{\partial D_c} = \frac{1}{\lambda_c}\sum_{u=1}^{h}\big(e^{\lambda_c u} - 1\big), \qquad \Delta_{L_c}T = T(L_c+1) - T(L_c).$$

With the additive residual model, the sensitivity to variable x_j is

$$\frac{\partial \widehat{T}}{\partial x_j} = \frac{W_j}{\sigma_j}f_j'(z_j) + \frac{\partial g_{\text{interaction}}}{\partial x_j}.$$

Proof. The first two derivatives are taken directly from the closed form of $S_c(n)$; the derivative with respect to P equals the number of pit stops because each pit stop adds exactly P. With the exponential cliff, D_c only multiplies linearly with the sum $\lambda_c^{-1}\sum_{u=1}^{h}(e^{\lambda_c u} - 1)$. The parameter L_c is discrete, so a finite difference is used. The residual formula comes from the chain rule with $z_j = (x_j - \mu_j)/\sigma_j$. □

Proposition 1.8.4 (Robustness checks)**.** An A6(c) evaluation is considered robust if the main conclusions are stable across a race-wise train-test split, leave-one-race-out, race-wise bootstrap, changing the seed, injecting small noise into the input variables and ablation by variable group. With weights W_j, stability can be measured by

$$\text{CV}(W_j) = \frac{\text{sd}(W_j)}{|\bar{W}_j| + 10^{-6}},$$

and the permutation importance of variable j is

$$I_j = \text{MAE}(X_{\pi(j)}) - \text{MAE}(X).$$

Proof. Each check changes one source of randomness or one part of the data but keeps the same evaluation protocol. If the variable ranking, the sign of the contribution and the test MAE vary little, the conclusion does not depend on one race, one seed or a single proxy. The CV formula normalizes the standard deviation by the mean magnitude of the weight; the I_j formula measures the loss when the information of variable j is destroyed, so it is a direct measure of its predictive role. □

2 A1. Single-Stint Time Model and Single-Compound Strategy

2.1 A1(a). Closed Form for $S_c(n)$ and the Total Time

If $n \le L_c$, i.e. the cliff has not yet been reached:

$$\boxed{S_c(n) = \sum_{1}^{n} (B_c + d_c(k-1)) = nB_c + \frac{d_c}{2} n(n-1)}$$

If $n > L_c$, i.e. the cliff has been passed, we can split it into the part before the cliff and the part after the cliff. The part before the cliff:

$$S_c(L_c) = \sum_{1}^{L_c} (B_c + d_c(k-1)) = L_c B_c + \frac{d_c}{2} L_c(L_c-1)$$

The part after the cliff: (B_c now becomes $B_c + d_c(L_c-1)$ because counting starts from level L_c rather than 1)

$$\sum_{k=L_c}^{n} \big(B_c + d_c(L_c-1) + D_c(k-1)\big)$$
$$= (n-L_c)\big(B_c + d_c(L_c-1)\big) + \frac{D_c}{2}(n-L_c)(n-L_c+1).$$

Adding them up, we obtain:

$$\boxed{S_c(n) = nB_c + \frac{d_c}{2}(L_c-1)(2n-L_c) + \frac{D_c}{2}(n-L_c)(n-L_c+1)}$$

For $n > L_H$, every choice of compound always passes the cliff since $L_3 > L_2 > L_1$, so the general formula is:

$$S_c(n) = nB_c + \frac{d_c}{2}(L_c-1)(2n-L_c) + \frac{D_c}{2}(n-L_c)(n-L_c+1), \qquad c \in \{S,M,H\}.$$

For $L_H \ge n > L_M$:

$$S_S(n) = nB_S + \frac{d_S}{2}(L_S-1)(2n-L_S) + \frac{D_S}{2}(n-L_S)(n-L_S+1)$$

$$S_M(n) = nB_M + \frac{d_M}{2}(L_M-1)(2n-L_M) + \frac{D_M}{2}(n-L_M)(n-L_M+1)$$

$$S_H(n) = nB_H + \frac{d_H}{2} n(n-1)$$

For $L_M \ge n > L_S$:

$$S_S(n) = nB_S + \frac{d_S}{2}(L_S-1)(2n-L_S) + \frac{D_S}{2}(n-L_S)(n-L_S+1)$$

$$S_M(n) = nB_M + \frac{d_M}{2} n(n-1)$$

$$S_H(n) = nB_H + \frac{d_H}{2} n(n-1)$$

For $L_S \ge n$:

$$S_c = nB_c + \frac{d_c}{2} n(n-1), \forall c \in \{S,M,H\}$$

Tire degradation model: lap time versus tire age (sample track)

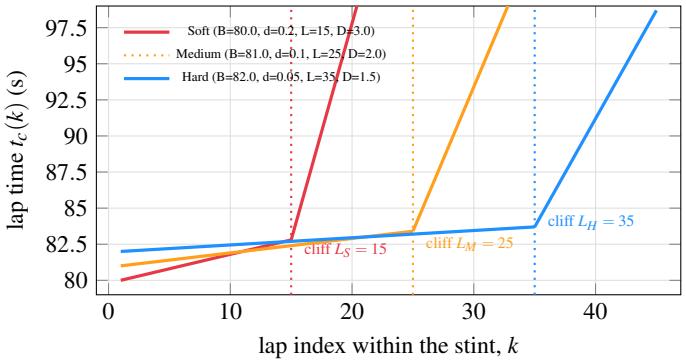


Figure 1: Tire degradation model on the sample track. The soft tire starts faster but degrades faster and reaches the cliff L_c earlier; after that the slope increases from d_c to D_c.

2.2 A1(b). Optimal Number of Pit Stops When Using Only One Compound

Since $L_c \in \mathbb{N}^*$, and by the convention above an F1 lap has only one pit stop, located just before the start line, when the cliff is reached the car can immediately enter the pit and change tires before passing the cliff.
In the sample data, the optimal solution for each compound does not need to extend the stint far beyond the cliff, because the large coefficient D_c makes the time increase sharply. However, in terms of the model, we must still allow $n > L_C$ and only exclude it when a proof or calculation shows it is not optimal. We can let the car run stints of length n (before the cliff) over the N laps. Then:

$$S_c = nB_c + \frac{d_c}{2} n(n-1)$$

If the N laps are covered by stints of length n, the number of stints is approximately:

$$m \approx \frac{N}{n}$$

For each stint, if one pit stop is amortized into that stint, the average cost per lap is approximately:

$$\bar{t}_c(n) = \frac{S_c(n) + P}{n} = B_c + \frac{d_c}{2}(n-1) + \frac{P}{n}$$

Note: the true total pit time of the race is $(m-1)P$, not mP. However, if $m \approx N/n$, then the true average pit cost is

$$\frac{(m-1)P}{N} = \frac{P}{n} - \frac{P}{N}$$

The term $-\dfrac{P}{N}$ is constant in n, so it does not change the location of the minimum. Therefore the formula

$$\frac{S_c(n) + P}{n}$$

can still be used to find a near-optimal stint length.
We take the derivative:

$$\frac{d\bar{t}_c}{dn} = \frac{d_c}{2} - \frac{P}{n^2}$$

The minimum occurs when:

$$\frac{d_c}{2} - \frac{P}{n^2} = 0$$

So the continuous optimal stint length is:

$$\boxed{n_c^* = \sqrt{\frac{2P}{d_c}}}$$

The corresponding number of stints is:

$$\boxed{m_c^* = \frac{N}{n_c^*} = N\sqrt{\frac{d_c}{2P}}}$$

The minimal average time, if not yet constrained by the cliff, is:

$$\boxed{\bar{t}_c^* = B_c - \frac{d_c}{2} + \sqrt{2Pd_c}}$$

Next, we prove that if $n_c^* > L_c$ then we choose $n_c^* = L_c$.
We need to prove that on the interval

$$1 \le n \le L_c$$

the function $\bar{t}_c(n)$ is decreasing, so its minimum is attained at $n = L_c$.

$$\forall n \le L_c\,,\, n < n_c^*$$
$$\Rightarrow n^2 < (n_c^*)^2 = \frac{2P}{d_c}$$

An equivalent transformation gives

$$\bar{t}_c'(n) = \frac{d_c}{2} - \frac{P}{n^2} < 0, \qquad \forall n \le L_c$$

Therefore, $\bar{t}_c(n)$ is decreasing on the entire interval before the cliff.

Hence, if constrained by $n \le L_c$, the minimum value of $\bar{t}_c(n)$ is attained at the largest point of the feasible region, i.e. $n^* := L_c$
If $n > L_c$ is allowed, we cannot rely only on the condition $n_c^* > L_c$ to conclude that the true optimal solution is L_c. After the cliff, the time formula changes. The first lap after L_c has marginal cost:

$$t_c(L_c+1) = B_c + d_c(L_c-1) + D_c$$

For L_c to truly be optimal compared with running one more lap, we need:

$$\frac{S_c(L_c+1)+P}{L_c+1} \ge \frac{S_c(L_c)+P}{L_c}$$
$$L_c\big(S_c(L_c+1)+P\big) \ge (L_c+1)\big(S_c(L_c)+P\big)$$
$$L_c\big(S_c(L_c)+t_c(L_c+1)+P\big) \ge (L_c+1)\big(S_c(L_c)+P\big)$$
$$L_c\,t_c(L_c+1) \ge S_c(L_c)+P$$
$$t_c(L_c+1) \ge \frac{S_c(L_c)+P}{L_c}$$
$$B_c + d_c(L_c-1) + D_c \ge B_c + \frac{d_c}{2}(L_c-1) + \frac{P}{L_c}$$
$$D_c \ge \frac{P}{L_c} - \frac{d_c}{2}(L_c-1)$$

If this condition holds, then already the first lap after the cliff has a marginal cost larger than the average cost at L_c. Computing with the data in the problem, we find that this holds; therefore if $n_c^* > L_c$ we choose $n_c^* = L_c$, $m_c^* = \lfloor N/L_C \rfloor + 0.5 \pm 0.5$
(i.e. we consider the 2 integers closest to the value N/L_C).

Next, we find the specific value of n when $n_c^* < L_c$. Suppose there are two stints i, j such that

$$n_i \ge n_j + 2$$

That is, stint i is longer than stint j by at least 2 laps.

We try moving 1 lap from the long stint i to the short stint j. Then

$$n_i \to n_i - 1 \quad \text{and} \quad n_j \to n_j + 1$$

so the total number of laps is unchanged.

The change in the pure running time is

$$\Delta = S_c(n_i-1) + S_c(n_j+1) - S_c(n_i) - S_c(n_j)$$

Using the identities

$$S_c(n_i-1) = S_c(n_i) - t_c(n_i)$$

and

$$S_c(n_j+1) = S_c(n_j) + t_c(n_j+1)$$

we obtain

$$\Delta = \big[S_c(n_i) - t_c(n_i)\big] + \big[S_c(n_j) + t_c(n_j+1)\big] - S_c(n_i) - S_c(n_j)$$
$$= -t_c(n_i) + t_c(n_j+1)$$

Since

$$n_i \ge n_j + 2$$

it follows that

$$n_j + 1 \le n_i - 1 < n_i$$

And since $t_c(n)$ increases with n,

$$t_c(n_j+1) \le t_c(n_i)$$

Therefore

$$\Delta = -t_c(n_i) + t_c(n_j+1) \le 0$$

This means that moving 1 lap from the long stint to the short stint does not increase the total time, and if $t_c(n)$ is strictly increasing it even decreases the total time.

Therefore, if a plan still has two stints differing by 2 laps or more, we can always create a better plan, or at least one that is no worse, by making them more balanced.

Repeating this move a finite number of times, eventually every pair of stints differs by at most one lap.

Now we can compute m^* and consider $m^* := \lceil m^* \rceil$ and $m^* := \lfloor m^* \rfloor$; then for each case we choose the n_i as close to each other as possible and compute the optimal case.

We thus obtain the following table:

Tire	$n^* = \sqrt{2P/d_c}$	$\lceil N/L_c \rceil$	optimal m	Strategy	T (s)
S	15.81 ($> L_S = 15$, locked by the cliff)	4	**4** (3 pits)	13-13-12-12	**4132.60**
M	22.36 ($\le L_M = 25$)	2	**2** (1 pit)	25-25	4135.00
H	31.62 ($\le L_H = 35$, but $m^* = 1.58$)	2	**2** (1 pit)	4155.00	

A1 — single-compound strategy: T versus number of stints ($N = 50,\ P = 25$)

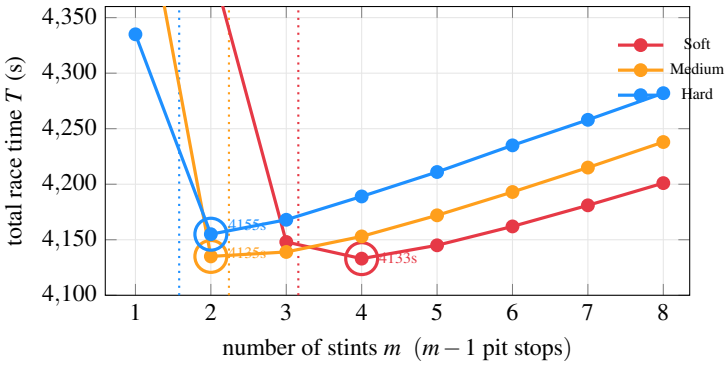


Figure 2: Illustration of the single-compound strategy in A1. The curves show the total time T versus the number of stints m; the dotted lines are the continuous solution $m^\star = N\sqrt{d_c/2P}$.

Conclusion of A1

For Soft and Hard, the continuous optimum m^* lies outside the region that avoids the cliff, so the solution is locked by the cliff constraint (Soft forces $m \ge 4$, Hard forces $m \ge 2$); only Medium has a feasible interior minimum. Among single-compound strategies, **Soft 3 pits (13–13–12–12),** $T \approx 4132.6$ s is the best, faster than Medium 1 pit (4135 s) and Hard 1 pit (4155 s).

3 A2. Optimal Strategy with Multiple Compounds

Lemma 1: Marginal cost balance
in an optimal solution, the last lap of any stint must not have a marginal cost greater than the cost of the next lap if it were added to another stint.

Consider a strategy with m stints, in which the compound of each stint is fixed:

$$(c_1, n_1), (c_2, n_2), \ldots, (c_m, n_m)$$

with

$$n_i \in \mathbb{Z}_{>0} \qquad \sum_{i=1}^{m} n_i = N$$

When the number of stints and the compounds are fixed, the number of pit stops is also fixed, so the pit time is constant. Therefore it suffices to minimize

$$\sum_{i=1}^{m} S_{c_i}(n_i)$$

Let the marginal cost of the last lap in stint i be

$$\Delta_i(n_i) = S_{c_i}(n_i) - S_{c_i}(n_i - 1) = t_{c_i}(n_i).$$

By definition:

$$S_c(n) = \sum_{k=1}^{n} t_c(k)$$

so

$$S_c(n) - S_c(n-1) = t_c(n)$$

Suppose there exist two stints i, j in an optimal plan such that

$$t_{c_i}(n_i) > t_{c_j}(n_j + 1)$$

Consider moving one lap from stint i to stint j:

$$n_i \mapsto n_i - 1, \qquad n_j \mapsto n_j + 1$$

Then

$$(n_i - 1) + (n_j + 1) = n_i + n_j$$

so the total number of laps is unchanged. At the same time, the number of stints and the compound of each stint are also unchanged, so the pit time does not change.

The change in the total running time is

$$\begin{aligned}\Delta &= S_{c_i}(n_i - 1) + S_{c_j}(n_j + 1) - S_{c_i}(n_i) - S_{c_j}(n_j).\\ &= \left[S_{c_i}(n_i) - t_{c_i}(n_i)\right] + \left[S_{c_j}(n_j) + t_{c_j}(n_j + 1)\right] - S_{c_i}(n_i) - S_{c_j}(n_j)\\ &= -t_{c_i}(n_i) + t_{c_j}(n_j + 1)\end{aligned}$$

By hypothesis $t_{c_i}(n_i) > t_{c_j}(n_j + 1)$ so $\Delta < 0$ This means that after moving one lap from stint i to stint j, the total time decreases, contradicting the assumption that the original plan is optimal.

Therefore there cannot exist any pair i, j such that

$$t_{c_i}(n_i) > t_{c_j}(n_j + 1)$$

Hence, in every interior optimal solution, we must have

$$t_{c_i}(n_i) \le t_{c_j}(n_j + 1), \qquad \forall i, j$$

Taking the largest left-hand side and the smallest right-hand side, we obtain

$$\max_i t_{c_i}(n_i) \le \min_j t_{c_j}(n_j + 1)$$

Meaning.

The condition above shows that in an optimal solution, the last lap of any stint must not have a marginal cost greater than the cost of the next lap if it were added to another stint. If there is a stint whose last lap is too expensive while another stint can still take one more lap at a lower cost, then moving the lap will reduce the total time. Therefore, the optimal state is exactly the state of marginal cost balance between the stints.

Lemma 2. Comparing compounds by stint

For a stint of n laps, let

$$g(n) = min\{S_S(n), S_M(n), S_H(n)\}$$

Soft

If $n \le L_S = 15$:

$$S_S(n) = 80n + 0.1n(n-1)$$

If $n > L_S = 15$:

$$S_S(n) = 1.5n^2 + 39.3n + 294$$

Medium

If $n \le L_M = 25$:

$$S_M(n) = 81n + 0.05n(n-1)$$

If $n > L_M = 25$:

$$S_M(n) = n^2 + 34.4n + 570$$

Hard

If $n \le L_H = 35$:

$$S_H(n) = 82n + 0.025n(n-1)$$

If $n > L_H = 35$:

$$S_S(n) = 0.75n^2 + 31.95n + 862.75$$

On the interval $[1, 15]$, none of the 3 tires has passed the cliff. Comparing:

$$S_S(n) - S_M(n) = n(0.05n - 10.5) < 0 \forall n \in [1, 15]$$

$$S_S(n) - S_H(n) = n(0.075n - 2.075) < 0 \forall n \in [1, 15]$$

So on the interval $[1, 15]$ we have $g(n) = S_S(n)$

On the interval $[15, 25]$, the Soft tire has passed the cliff while the other 2 tires have not. Comparing:

$$\begin{aligned}S_S(n) - S_M(n) &= 1.45n^2 - 41.65n + 294,\\ &< 0 \quad (n = 16), \qquad > 0 \quad (17 \le n \le 25).\end{aligned}$$

$$S_M(n) - S_H(n) = n(0.025n - 1.025) < 0 \forall n \in [16, 25]$$

So at $n = 16$ we have $g(16) = S_S(16)$ and on the interval $[17, 25]$ we have $g(n) = S_M(n)$

On the interval $[25, 35]$, Soft and Medium have passed the cliff while Hard has not. Comparing:

$$S_S(n) - S_M(n) = 0.5n^2 + 4.9n - 276 > 0 \forall n \in [26, 35]$$

$$\begin{aligned}S_M(n) - S_H(n) &= 0.975n^2 - 47.575n + 570,\\ &< 0 \quad (n \in \{26, 27\}), \qquad > 0 \quad (28 \le n \le 35).\end{aligned}$$

So at $n = 26$ or 27 we have $g(n) = S_M(n)$ and on the interval $[28, 35]$ we have $g(n) = S_H(n)$

On the interval $[36, 50]$, all 3 tires have passed the cliff. Comparing:

$$S_S(n) - S_H(n) = 0.75n^2 + 7.35n - 568.75 > 0 \forall n \in [36, 50]$$

$$S_M(n) - S_H(n) = 0.25n^2 + 2.45n - 292.75 > 0 \forall n \in [36, 50]$$

So on the interval $[35, 50]$ we have $g(n) = S_H(n)$

Therefore:

$$g(n) = \begin{cases} S_S(n) = 80n + 0.1n(n-1), & 1 \le n \le 15 \\ S_S(n) = 1.5n^2 + 39.3n + 294, & n = 16 \\ S_M(n) = 81n + 0.05n(n-1), & 17 \le n \le 25 \\ S_M(n) = n^2 + 34.4n + 570, & 26 \le n \le 27 \\ S_H(n) = 82n + 0.025n(n-1), & 28 \le n \le 35 \\ S_H(n) = 0.75n^2 + 31.95n + 862.75, & 36 \le n \le 50 \end{cases}$$

Next, for each compound we need to compute how many stints are optimal.

Soft

The continuous optimal stint length is

$$n_S^* = \sqrt{\frac{2P}{d_S}} = \sqrt{\frac{50}{0.2}} \approx 15.81$$

But since $15.81 > L_S = 15$, the continuous solution passes the cliff. By the result of A1, the cliff locks the solution, so we choose

$$\boxed{n_S = 15}$$

The time of one stint is

$$S_S(15) = 15 \cdot 80 + \frac{0.2 \cdot 15 \cdot 14}{2} = 1221$$

The average cost per lap (with the pit amortized) is

$$\bar{t}_S(15) = \frac{S_S(15)+P}{15} = \frac{1221+25}{15} = 83.0667$$

Therefore Soft is most economical when run for

$$\boxed{\text{15 laps/stint}}$$

Medium The continuous optimal stint length is

$$n_M^* = \sqrt{\frac{2P}{d_M}} = \sqrt{\frac{50}{0.1}} \approx 22.36$$

Since $22.36 < L_M = 25$, the optimal solution lies before the cliff.
Check the two nearest integers, 22 and 23:

$$S_M(22) = 22 \cdot 81 + \frac{0.1 \cdot 22 \cdot 21}{2} = 1805.1$$

$$\bar{t}_M(22) = \frac{1805.1+25}{22} = 83.1864$$

and

$$S_M(23) = 23 \cdot 81 + \frac{0.1 \cdot 23 \cdot 22}{2} = 1888.3$$

$$\bar{t}_M(23) = \frac{1888.3+25}{23} = 83.187$$

Therefore

$$\boxed{n_M = 22}$$

So Medium is most economical when run for about

$$\boxed{\text{22 laps/stint}}$$

Hard The continuous optimal stint length is

$$n_H^* = \sqrt{\frac{2P}{d_H}} = \sqrt{\frac{50}{0.05}} \approx 31.62$$

Since $31.62 < L_H = 35$, the optimal solution is not yet locked by the cliff.
Check the nearest integers, 31 and 32:

$$S_H(32) = 32 \cdot 82 + \frac{0.05 \cdot 32 \cdot 31}{2} = 2648.8$$

$$\bar{t}_H(32) = \frac{2648.8+25}{32} = 83.5563$$

and

$$S_H(31) = 31 \cdot 82 + \frac{0.05 \cdot 31 \cdot 30}{2} = 2565.25$$

$$\bar{t}_H(31) = \frac{2565.25+25}{31} = 83.5565$$

Therefore

$$\boxed{n_H = 32}$$

So Hard is most economical when run for about

$$\boxed{\text{32 laps/stint}}$$

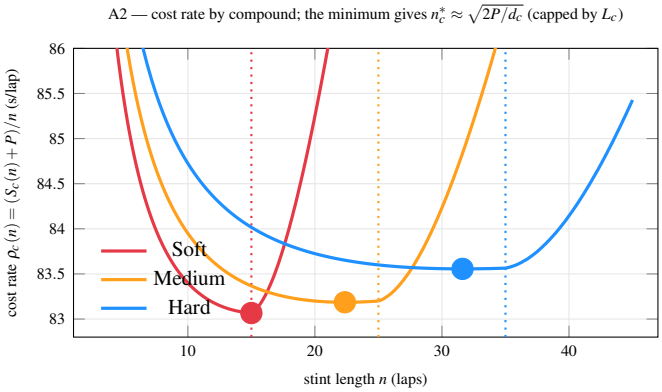


Figure 3: Average cost rate versus stint length for each compound. The dots mark the near-optimal stint length; the vertical dotted lines show the cliff L_c.

Comparison of the Three Economical Regimes

Tire	Economical stint length	$S_c(n)$	$\frac{S_c(n)+P}{n}$
Soft	15	1221.0	83.0667
Medium	22	1805.1	83.1864
Hard	32	2648.8	83.5563

The key conclusion is

$$\boxed{\bar{t}_S(15) < \bar{t}_M(22) < \bar{t}_H(32)}$$

Therefore, Soft with a stint length of about 15 laps is the most efficient running unit.

Hence, in problem A2, a reasonable strategy is to use Soft stints of length close to 15 laps as the backbone. In other words, we want to exploit Soft as much as possible but without extending it past the cliff, because after lap 15 the marginal cost increases significantly.

We use k Soft stints of 15 laps (denoted S15) to start the race, until the number of remaining laps drops to the level L_H. When it drops to L_H, we have the choice of using the Hard tire for the rest of the race or continuing to use Soft together with another tire combination.

According to the problem, after running S15 we have $N := 50 - 15 = 35$

Using S15+H35
The time to run Hard is:

$$S_H = 2899.75$$

Total time used:

$$T = S_S(15) + S_H(35) + P = 4145.75$$

Using S15+S15+...
Then 20 laps remain and we can use a combination of 2 or 3 S15. The options are to run M20, H20, S15+M5, S15+H5, S15+S5

Consider the 2 functions $t_M(k) = 81 + 0.1(k-1)$ and $t_H(k) = 82 + 0.05(k-1)$:

On the range $\{1,21\}$ we have $t_M(k) \le t_H(k)$ and on $\{21,25\}$ we have $t_M(k) \ge t_H(k)$ Since $20 < 21$, in this range Hard is larger than Medium, so we can eliminate Hard. Similarly, S5 is better than M5. So M20 and S15+S5 remain.

- Using S15+S15+M20:
  Apply Lemma 1 to prove optimality:

$$t_S(15) \le t_M(21)$$

$$t_M(20) \le t_S(16)$$

  so there is no way to move a lap between the stints. The calculation gives $T = 4131$s.
- Using S15+S15+S15+S5:
  We see that

$$S_M(20)1639 < 1648 = S_S(15) + 25 + S_S(5)$$

  so this option is not as good as the one above.

The options are to run M20, H20, S15+M5, S15+H5, S15+S5

From the average-cost analysis, we see that a Soft stint of about 15 laps is the most efficient running unit, because Soft has the smallest base time and has not yet passed the cliff $L_S = 15$. Therefore, the optimal strategy should use 15-lap Soft stints as the backbone. However, if too many Soft stints are used, an extra pit stop arises for a remaining segment that is too short, losing the time advantage.

Table 1: Comparison of the three candidate strategies in A2

Strategy	Number of stints	Running time	Pit time	Total time
$S15+H35$	2	4120.75	25	4145.75
S15 + S15 + M20	**3**	**4081.00**	**50**	**4131.00**
$S15+S15+S15+S5$	4	4065.00	75	4140.00

Conclusion of Task 2

Therefore the optimal approach is

$$\boxed{S15+S15+M20}$$

with the smallest total time of 4131 seconds.

4 A3. Strategy Analysis by Number of Pit Stops

Let A_m be the minimal running time to complete 50 laps in m stints, not counting the pit-stop time:

$$T_m(P) = A_m + (m-1)P$$

Since we may choose any number of stints, the optimal time is:

$$T^*(P) = \min_{1\le m\le 50} T_m(P)$$

One can picture A_m as a lower envelope of the lines $T_m(P)$. The strategy changes at the positions where two or more lines intersect.

Lemma the optimal number of stints does not increase as P increases
Suppose $P_2 > P_1$. Let m_1 be the optimal number of stints for P_1 and m_2 the optimal number of stints for P_2. By optimality:

$$T_{m_1}(P_1) \le T_{m_2}(P_1)$$

$$T_{m_2}(P_2) \le T_{m_1}(P_2)$$

Adding side by side:

$$\begin{aligned} &A_{m_1} + (m_1-1)P_1 + A_{m_2} + (m_2-1)P_2 \\ &\quad \le A_{m_2} + (m_2-1)P_1 + A_{m_1} + (m_1-1)P_2. \end{aligned}$$

$$(m_2-m_1)(P_2-P_1) \le 0$$

Since $P_2 > P_1$, we get $m_2 \le m_1$. Hence, when the pit time increases, the number of stints stays the same or decreases. That is, as P increases, m may follow a decreasing staircase or stay unchanged.

The next goal is to prove that for every $m \ge 4$, the optimal strategy uses only the Soft tire with lengths that are as balanced as possible.

Marginal cost $\delta(n)$
Define

$$\delta(n) = g(n) - g(n-1)$$

This is the additional time when the optimal length of a stint is increased from $n-1$ to n laps. At the points where the compound changes, $\delta(n)$ is the difference of two values on the envelope g, so it must be computed exactly from the definition of $g(n)$, and must not automatically be taken as the n-th lap of the same compound.

Idea: we want to prove that if there is a stint longer than 16 laps, then there exists a stint shorter than 12 laps. Then we can move 1 lap from the long stint to the short stint and reduce the time.

Step 1: compute $\max_{1\le n\le 12}\delta(n)$
For $1 \le n \le 12$, the Soft tire is the cheapest and has not yet reached the sharp-degradation threshold. Therefore:

$$\begin{aligned} \delta(n) &= g(n) - g(n-1) \\ &= S_{\mathrm{S}}(n) - S_{\mathrm{S}}(n-1) \\ &= 80 + 0.2(n-1) \end{aligned}$$

$$\Rightarrow \max_{1\le n\le 12}\delta(n) = \delta(12) = 82.2$$

Step 2: compute $\min_{17\le n\le 50}\delta(n)$
We must consider the points $n = 17$ and $n = 28$ separately, because at those points the compound with the least time changes.

Case 1: $n = 17$

$$\delta(17) = g(17) - g(16) = S_M(17) - S_S(16) = 1390.6 - 1306.8 = 83.8$$

Case 2: $18 \le n \le 25$**.** Both $g(n)$ and $g(n-1)$ use Medium, so

$$\delta(n) = S_{\mathrm{M}}(n) - S_{\mathrm{M}}(n-1) = 81 + 0.1(n-1) \ge 82.7$$

Equality holds at $n = 18$. Specifically:

$$\delta(18) = g(18) - g(17) = S_{\mathrm{M}}(18) - S_{\mathrm{M}}(17) = 1473.3 - 1390.6 = 82.7.$$

Case 3: $26 \le n \le 27$ Medium has passed the 25-lap threshold, so

$$\delta(n) = 83.4 + 2(n-25) \ge 85.4$$

Case 4: $n = 28$

$$\delta(28) = g(28) - g(27) = S_H(28) - S_M(27) = 2314.9 - 2227.8 = 87.1$$

Case 5: $29 \le n \le 35$
Both consecutive values use Hard before the threshold, so

$$\delta(n) = 82 + 0.05(n-1) \ge 83.4$$

Case 6: $36 \le n \le 50$
Hard has passed the 35-lap threshold, so

$$\delta(n) = 83.7 + 1.5(n-35) \ge 85.2$$

The smallest value among all the cases above is 82.7, attained at $n = 18$. Therefore

$$\boxed{\min_{17\le n\le 50}\delta(n) = \delta(18) = 82.7}$$

Lap-transfer argument
Suppose an optimal plan has $m \ge 4$ stints and has a stint of length $a \ge 17$ laps. Then there must exist another stint of length $b \le 11$ laps. Indeed, if all the remaining stints had length at least 12 laps then

$$50 = \sum_{i=1}^{m} n_i \ge 17 + 12(m-1) \ge 17 + 12\cdot 3 = 53$$

a contradiction.

Now move one lap from the long stint a to the short stint b. The two lengths change from (a,b) to $(a-1, b+1)$. The change in running time is

$$\begin{aligned} \Delta_{\text{transfer}} &= \big[g(a-1) + g(b+1)\big] - \big[g(a) + g(b)\big] \\ &= \big[g(b+1) - g(b)\big] - \big[g(a) - g(a-1)\big] \\ &= \delta(b+1) - \delta(a) \end{aligned}$$

Since $b \le 11$, we have $b+1 \le 12$, so:

$$\delta(b+1) \le 82.2$$

Since $a \ge 17$:

$$\delta(a) \ge 82.7$$

Hence

$$\Delta_{\text{transfer}} = \delta(b+1) - \delta(a) \le 82.2 - 82.7 = -0.5$$

The negative sign means that the new plan is faster by at least 0.5 seconds, contradicting the assumption that the original plan is optimal.

Conclusion: In every optimal strategy with $m \ge 4$ stints, no stint has length 17 laps or more. In other words, every length is at most 16 laps.

For every length from 1 to 16 laps, the Soft tire has the smallest total time. Therefore, when $m \ge 4$, every optimal stint uses Soft.

With the same Soft tire, if two lengths satisfy $a \ge b+2$, moving one lap from the long stint to the short stint changes the time by

$$\Delta t = \big[S_S(a-1) + S_S(b+1)\big] - \big[S_S(a) + S_S(b)\big] = t_S(b+1) - t_S(a) < 0$$

because the lap time $t_S(k)$ increases with tire age. Repeating this move until no two lengths differ by 2 laps or more, we obtain a balanced configuration.

We compute q, the base length of a stint when m is fixed:

$$q = \left\lfloor \frac{50}{m} \right\rfloor$$

Then $r = 50 - mq$ laps are still missing, so these r laps must be made up by $q+1$, i.e. there are then r stints of length $q+1$ laps and $m-r$ stints of length q laps. For $m \geq 4$:

$$q+1 \leq \left\lceil \frac{50}{4} \right\rceil = 13$$

so the balanced stints all lie before the sharp-degradation threshold of Soft. Therefore

$$\begin{aligned} A_m &= (m-r)S_S(q) + rS_S(q+1) \\ &= (m-r)\big[80q + 0.1q(q-1)\big] + r\big[80(q+1) + 0.1q(q+1)\big] \\ &= 80\big[(m-r)q + r(q+1)\big] + 0.1\big[(m-r)q(q-1) + rq(q+1)\big] \\ &= 80(50) + 0.1\big[mq(q-1) + 2rq\big] \\ &= 4000 + 0.1\big[mq(q-1) + 2rq\big], \qquad m \geq 4 \end{aligned}$$

For example, with $m = 4$ we have $q = 12$, $r = 2$, so

$$A_4 = 4000 + 0.1\big[4(12)(11) + 2(2)(12)\big] = 4057.6$$

The corresponding strategy is $2S_S(13) + 2S_S(12)$.

Marginal value of an additional stint

Define

$$\Delta_m = A_m - A_{m+1}$$

Δ_m is the number of seconds of tire running time saved when increasing from m to $m+1$ stints.

Comparing the two total times directly:

$$\begin{aligned} T_{m+1}(P) - T_m(P) &= \big[A_{m+1} + mP\big] - \big[A_m + (m-1)P\big] \\ &= P - (A_m - A_{m+1}) \\ &= P - \Delta_m \end{aligned}$$

Therefore,

$\Delta_m > P$	$\iff T_{m+1}(P) < T_m(P)$	(add a stint)
$\Delta_m < P$	$\iff T_{m+1}(P) > T_m(P)$	(do not add)
$\Delta_m = P$	$\iff T_{m+1}(P) = T_m(P)$	(the two choices tie)

We build a table of marginal benefits to show when one more stint should be added:

m	$\Delta_m = A_m - A_{m+1}$
1	225.25
2	29.0
3	23.4
4	12.6
5	8.2
6	6.0
7	4.4
8	3.4
9	3.0
10–11	2.0
12	1.6
13–15	1.2
16	1.0
17–24	0.6
25–49	0.2

The sequence Δ_m does not increase as there are more stints. Hence, for a P that does not lie exactly at a threshold, the optimal number of stints m is identified by the two conditions

$$\Delta_{m-1} > P > \Delta_m$$

The left condition says that going from $m-1$ to m is still beneficial, and the right condition says that going further from m to $m+1$ is no longer beneficial.

All strategy-change thresholds

Applying the method above to the sequence Δ_m, we obtain the following complete table. The order of the stints can be swapped without changing the total time in the model.

Range of P (seconds)	m	Pits	Optimal stint set	$T^*(P)$
$0 \leq P < 0.2$	50	49	$50\times$S1	$4000+49P$
$0.2 < P < 0.6$	25	24	$25\times$S2	$4005+24P$
$0.6 < P < 1.0$	17	16	$16\times$S3$+$S2	$4009.8+16P$
$1.0 < P < 1.2$	16	15	$2\times$S4$+14\times$S3	$4010.8+15P$
$1.2 < P < 1.6$	13	12	$11\times$S4$+2\times$S3	$4014.4+12P$
$1.6 < P < 2.0$	12	11	$2\times$S5$+10\times$S4	$4016.0+11P$
$2.0 < P < 3.0$	10	9	$10\times$S5	$4020.0+9P$
$3.0 < P < 3.4$	9	8	$5\times$S6$+4\times$S5	$4023.0+8P$
$3.4 < P < 4.4$	8	7	$2\times$S7$+6\times$S6	$4026.4+7P$
$4.4 < P < 6.0$	7	6	S8$+6\times$S7	$4030.8+6P$
$6.0 < P < 8.2$	6	5	$2\times$S9$+4\times$S8	$4036.8+5P$
$8.2 < P < 12.6$	5	4	$5\times$S10	$4045.0+4P$
$12.6 < P < 23.4$	4	3	$2\times$S13$+2\times$S12	$4057.6+3P$
$23.4 < P < 29.0$	3	2	$2\times$S15$+$M20	$4081.0+2P$
$29.0 < P < 225.25$	2	1	$2\times$M25	$4110.0+P$
$P > 225.25$	1	0	H50	4335.25

At each boundary value, the strategies in the two adjacent ranges have the same total time. Four thresholds have more than two stint counts that are jointly optimal because some consecutive values of Δ_m are equal:

Threshold P	Jointly optimal stint counts
0.2	$25, 26, \ldots, 50$
0.6	$17, 18, \ldots, 25$
1.2	$13, 14, 15, 16$
2.0	$10, 11, 12$

At every remaining threshold, only the two stint counts on either side are jointly optimal.

$$A3 \text{ — } T^*(P) = \min_m \big[B(m) + (m-1)P\big]$$

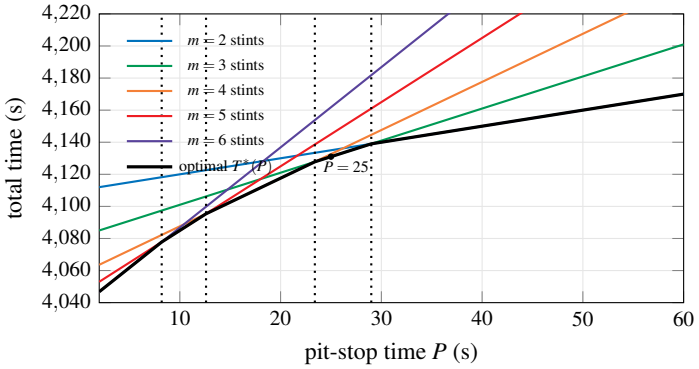


Figure 4: Illustration of the optimal function $T^*(P)$ versus the pit-stop time P. The colored lines are the linear functions for each number of stints; the black line is the lower envelope, i.e. the optimal strategy.

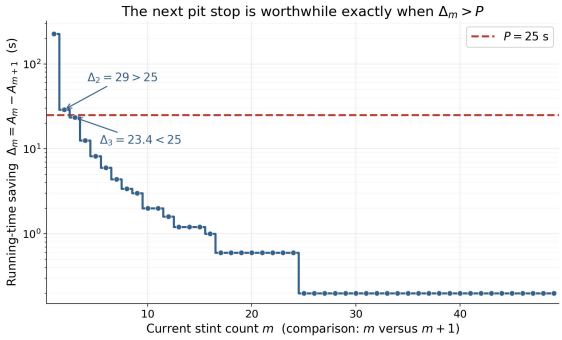


Figure 5: Additional pit condition: a new stint is worthwhile only when the marginal saving Δ_m is larger than the pit cost P.

Conclusion of A3

As P increases, the optimal number of stints decreases stepwise at the thresholds $P = \Delta_m$. With $P = 25$, $23.4 < P < 29$, so the optimal strategy is $S15 + S15 + M20$ with $T = 4131$ seconds.

5 A4. Structure of Candidate Strategies

5.1 Objective of A4

In A4, for each track in the 20-track dataset, we need to find a valid pit-stop strategy with the smallest total time.

A valid strategy is a sequence of stints

$$(c_1, n_1), (c_2, n_2), \ldots, (c_m, n_m)$$

where

$$c_i \in \{S, M, H\}, \qquad n_i \in \mathbb{Z}_{>0}, \qquad \sum_{i=1}^{m} n_i = N,$$

and the strategy must use at least two different tire compounds.

The objective is to minimize the total race time

$$T = \sum_{i=1}^{m} S_{c_i}(n_i) + (m-1)P$$

The main idea of the solution is: instead of directly comparing strategies with many stints, we construct an "ideal stint" standard via the average time per lap, and then turn the problem into minimizing the total penalty relative to that ideal standard.

5.2 Normalization by the Average Time per Lap

For each compound c and stint length n, let

$$C_c(n) = S_c(n) + P$$

Here $C_c(n)$ is the cost of a stint if we "assign" an additional pit cost P to that stint itself.

For a strategy with m stints, we have

$$\begin{aligned}\sum_{i=1}^{m} C_{c_i}(n_i) &= \sum_{i=1}^{m} \left[S_{c_i}(n_i) + P\right] \\ &= \sum_{i=1}^{m} S_{c_i}(n_i) + mP\end{aligned}$$

Meanwhile the true total time is

$$\begin{aligned}T &= \sum_{i=1}^{m} S_{c_i}(n_i) + (m-1)P \\ &= \sum_{i=1}^{m} C_{c_i}(n_i) - P\end{aligned}$$

Since P is a constant of the track, minimizing T is equivalent to minimizing

$$\sum_{i=1}^{m} C_{c_i}(n_i)$$

We have the average time per lap of the stint (c,n):

$$\bar{t}_c(n) = \frac{C_c(n)}{n} = \frac{S_c(n)+P}{n}$$

Let

$$\lambda = \min_{\substack{c\in\{S,M,H\} \\ 1\le n\le N}} \bar{t}_c(n)$$

be the ideal average time per lap.

Suppose the minimum is attained at

$$(c_0, n_0)$$

that is

$$\lambda = \bar{t}_{c_0}(n_0)$$

Then the stint (c_0,n_0) is called the most ideal stint.

5.3 Penalty Function Relative to the Ideal Stint

For each compound c and stint length n, define the penalty function

$$p_c(n) = C_c(n) - n\lambda = S_c(n) + P - n\lambda$$

The meaning of $p_c(n)$ is: if we run the stint (c,n), how many seconds the true cost of that stint exceeds the ideal standard $n\lambda$.

Since λ is the minimum value of $\bar{t}_c(n) = C_c(n)/n$, for all c,n,

$$\begin{aligned}\frac{C_c(n)}{n} &\ge \lambda \\ C_c(n) &\ge n\lambda \\ p_c(n) = C_c(n) - n\lambda &\ge 0\end{aligned}$$

For the ideal stint (c_0,n_0):

$$p_{c_0}(n_0) = C_{c_0}(n_0) - n_0\lambda = 0$$

Thus the ideal stint has a penalty equal to 0, while every other stint has a nonnegative penalty.

5.4 Lemma 1

Consider an arbitrary strategy

$$(c_1,n_1),(c_2,n_2),\ldots,(c_m,n_m)$$

satisfying

$$\sum_{i=1}^{m} n_i = N$$

We have

$$\begin{aligned}\sum_{i=1}^{m} C_{c_i}(n_i) &= \sum_{i=1}^{m} \left[n_i\lambda + p_{c_i}(n_i)\right] \\ &= \lambda \sum_{i=1}^{m} n_i + \sum_{i=1}^{m} p_{c_i}(n_i) \\ &= N\lambda + \sum_{i=1}^{m} p_{c_i}(n_i)\end{aligned}$$

Since

$$T = \sum_{i=1}^{m} C_{c_i}(n_i) - P$$

it follows that

$$T = N\lambda - P + \sum_{i=1}^{m} p_{c_i}(n_i)$$

For a fixed track, $N\lambda - P$ is a constant. Therefore minimizing T is equivalent to minimizing

$$R = \sum_{i=1}^{m} p_{c_i}(n_i)$$

Here R is the total penalty of the whole strategy.

Meaning.. This lemma allows us to convert the problem of optimizing race time into the problem of optimizing the total deviation from an ideal stint.

Without the constraint of using at least two compounds, we would simply want to use as many ideal stints as possible. But A4 requires using at least two compounds, so the problem becomes adding another compound so that the total penalty increases the least.

5.5 The Most Ideal and the Second Most Ideal Tire

The most ideal stint is

$$(c_0,n_0) = \arg\min_{c,n} \bar{t}_c(n)$$

For the two remaining compounds, we consider the smallest penalty that each compound can produce. Specifically, for each $c \neq c_0$ let

$$n_c^\star = \arg\min_{1\le n\le N} p_c(n)$$

Then $p_c(n_c^\star)$ is the smallest penalty if we are forced to use tire c.

We choose the second most ideal tire to be the compound

$$c_1 = \arg\min_{c\neq c_0} p_c(n_c^\star)$$

and set $n_1 = n_{c_1}^\star$

Thus we have two base stints:

$$C_L = (c_0,n_0)$$

is the most ideal stint, and

$$C_A = (c_1,n_1)$$

is the second most ideal stint.

Here

$$p_{c_0}(n_0) = 0$$

and

$$p_{c_1}(n_1)$$

is the smallest penalty needed to satisfy the requirement of using another compound.

5.6 Structure of the Candidate Strategies

Since the most ideal stint has a penalty equal to 0, the optimal strategy should use many stints of type C_L. However, since A4 requires using at least two compounds, the strategy must also have at least one stint of another type. The other type with the smallest penalty is exactly C_A.

The number of stints m is chosen close to the ratio

$$\frac{N}{n_0}$$

Specifically, we consider the two nearest integer values:

$$m \in \left\{ \left\lfloor \frac{N}{n_0} \right\rfloor, \left\lceil \frac{N}{n_0} \right\rceil \right\}$$

We consider candidate strategies of the form

$$xC_L + yC_A$$

where

$$x, y \in \mathbb{Z}_{\geq 1}, \qquad x \geq y, \qquad x + y = m$$

The condition $x \geq y$ reflects the optimality intuition: we should use more of the most ideal stint than of the second most ideal stint, because the most ideal stint has a smaller or equal penalty.

For each m, we consider the pairs (x, y) satisfying the conditions above.

The initial total number of laps of this combination is

$$L_0 = xn_0 + yn_1$$

If

$$L_0 \geq N$$

then this combination can be used after shortening some stints so that the total number of laps equals exactly N.

Let the number of excess laps be

$$e = L_0 - N$$

We need to shorten a total of exactly e laps from the stints in the combination $xC_L + yC_A$.

Suppose a stint currently has the form (c, r), i.e. it uses tire c for r laps.

If we shorten this stint from r laps to $r-1$ laps, the penalty changes from $p_c(r)$ to $p_c(r-1)$

If $\Delta_c(r)$ is small, this shortening operation increases the total penalty only slightly. If $\Delta_c(r)$ is large, this shortening operation makes the strategy considerably worse.

We have:

$$\begin{aligned} \Delta_c(r) &= \big[S_c(r-1) + P - (r-1)\lambda\big] - \big[S_c(r) + P - r\lambda\big] \\ &= \lambda - \big[S_c(r) - S_c(r-1)\big] \\ &= \Delta_c(r) = \lambda - t_c(r) \end{aligned}$$

When shortening a stint from r to $r-1$, we remove the r-th lap. The removed lap has time $t_c(r)$. Compared with the ideal standard λ, the change in penalty is exactly

$$\lambda - t_c(r)$$

In the present problem, we need to shorten a total of e laps. Therefore, at each step, we choose a shortening operation with the smallest $\Delta_c(r)$ among all current stints. In other words, we remove laps from the stint where shortening does the least damage to the total penalty.

Since the lap time $t_c(k)$ does not decrease in k, for the same compound c:

$$t_c(1) \leq t_c(2) \leq \cdots \leq t_c(r)$$

Therefore:

$$\Delta_c(r-1) = \lambda - t_c(r-1) \geq \lambda - t_c(r) = \Delta_c(r)$$

Therefore, for each stint, the sequence of marginal costs of successive shortening operations has an increasing structure.

When we need to choose a total of e shortening operations, we must choose a set of operations such that if a stint is shortened by several laps, the earlier shortening operations of that same stint have also been chosen. This is a "prefix"-type constraint.

Since the marginal costs within each stint are increasing, the following greedy algorithm is optimal:

1. At each step, consider all currently available shortening operations.
2. Choose the operation with the smallest marginal cost Δ.
3. Perform that operation.
4. Update the next marginal cost of the stint that was just shortened.
5. Repeat until exactly e laps have been removed.

Suppose there is an optimal shortening plan that at some step does not choose the available operation with the smallest Δ, but instead chooses an operation with a larger Δ. Replacing the operation with the larger Δ by the operation with the smaller Δ leaves the total number of removed laps unchanged, keeps the prefix constraint, and does not increase the total penalty. Hence there always exists an optimal plan that chooses the operation with the smallest Δ at that step. Repeating this argument for each step, the greedy algorithm is optimal.

The implementation has very small complexity. For each track, we compute the table of $S_c(n)$, $a_c(n)$, $p_c(n)$ for $3N$ values. Then we only consider a small number of combinations (m, x, y), and each combination only needs to be shortened by at most $O(N)$ laps. Since $N \leq 78$ in the dataset, the algorithm runs almost instantly.

If

$$L_0 < N$$

then we can do the same thing, but this time instead of decreasing we increase, and finally we again compare with all of them to choose the optimal model.

5.7 Numerical Results for the 20 Tracks

Table 2: Optimal A4 strategies for the 20 tracks. The "gap" column is the time difference to the next-best strategy structure.

ID	Track	Execution order	Pit after lap	Pits	Total time (s)	Gap (s)
1	Bahrain	M16+M16+M15+S10	16, 32, 47	3	5592.3750	0.4075
2	Jeddah	H28+M22	28	1	4603.5000	1.5000
3	Melbourne	M19+S13+S13+S13	19, 32, 45	3	4715.2950	1.1850
4	Suzuka	M17+S12+S12+S12	17, 29, 41	3	4919.5000	3.0000
5	Imola	H37+M26	37	1	4949.5100	2.4450
6	Monaco	H32+M23+M23	32, 55	2	5769.7800	1.8500
7	Barcelona	M19+M19+S14+S14	19, 38, 52	3	5190.8600	0.1000
8	Montreal	H28+M21+M21	28, 49	2	5274.6550	1.8775
9	Spielberg	M20+M20+M19+S12	20, 40, 59	3	4807.1600	2.0400
10	Silverstone	M20+M20+S12	20, 40	2	4733.4400	4.4800
11	Hungaroring	H22+M16+M16+M16	22, 38, 54	3	5581.5925	0.3375
12	Spa	M16+S14+S14	16, 30	2	4686.0000	6.2000
13	Zandvoort	H22+M17+M17+M16	22, 39, 56	3	5303.7750	1.7750
14	Monza	H31+M22	31	1	4339.3720	2.8840
15	Singapore	H24+M19+M19	24, 43	2	6079.2000	3.5200
16	COTA	M20+S12+S12+S12	20, 32, 44	3	5536.2500	0.5500
17	Mexico	M20+M19+M19+S13	20, 39, 58	3	5716.8400	0.5750
18	Interlagos	M20+M19+M19+S13	20, 39, 58	3	5238.5200	2.1250
19	LasVegas	H28+M22	28	1	4802.5000	0.5000
20	AbuDhabi	M22+M22+S14	22, 44	2	5248.8200	0.4750

Example: Bahrain

A detailed solution to derive, with justification, that the optimal strategy is

$$\boxed{\text{M16} \to \text{M16} \to \text{M15} \to \text{S10}}$$

5.8 Data and Time Model

The Bahrain data in Appendix A of the problem statement:

Tire c	B_c	d_c	L_c	D_c
Soft (S)	95.0	0.29	10	3.0
Medium (M)	95.9	0.145	17	2.0
Hard (H)	96.8	0.0725	24	1.5

$N =$ 57 laps
$P = 24$ seconds

The k-th lap on a set of tires c has time

$$t_c(k) = \begin{cases} B_c + d_c(k-1), & k \le L_c \\ B_c + d_c(L_c - 1) + D_c(k - L_c), & k > L_c \end{cases}$$

Total time:

$$T = \sum_{i=1}^{m} S_{c_i}(n_i) + (m-1)P$$

The three economical minima for Bahrain are

Package	$S_c(n)$	$A_c(n) = S_c(n) + 24$	$a_c(n)$
S10	963.050	987.050	98.705000
M17	1650.020	1674.020	**98.471765**
H24	2343.210	2367.210	98.633750

Taking the best average cost as the standard:

$$\lambda = \min_{c,n} a_c(n) = a_M(17) = 98.4717647 \text{ seconds/lap}$$

We compute

$$\rho_S(10) = 963.05 + 24 - 10\lambda = 2.332353$$

$$\rho_H(24) = 2343.21 + 24 - 24\lambda = 3.887647$$

Although $a_H(24) < a_S(10)$, the Hard stint is as long as 24 laps, so the small difference per lap accumulates into a larger excess. To satisfy the rule of using a second compound, $S10$ "pays" less than $H24$.

We obtain $m \approx N/n_0 = 57/17 \approx 3.35$, so $m = 3$ or $m = 4$. Note that if $m = 3$, the only combination $(x, y) = (2, 1)$ does not work because $2.17 + 10 < 57$. Therefore we take $m = 4$. Taking the combination $(x, y) = (3, 1)$, the initial model is:

$$S10 + M17 + M17 + M17$$

which has 61 laps, 4 more than required.

5.8.1 Which laps are cheapest to remove?

When reducing the package (c, n) to $(c, n-1)$, the penalty increases by

$$\rho_c(n-1) - \rho_c(n) = \lambda - t_c(n)$$

The relevant options are

Change	$t_c(n)$	penalty increase
$M17 \to M16$	98.220	0.251765
$M16 \to M15$	98.075	0.396765
$S10 \to S9$	97.610	0.861765

Since removing laps from Medium is cheaper than removing laps from Soft, we keep $S10$ and remove 4 laps from the three Medium stints: three times $17 \to 16$, then once $16 \to 15$.

The optimal stints have not passed the cliff, so we use the arithmetic-series formula to compute the total time:

$$S_S(10) = 10(95) + \frac{0.29 \cdot 10 \cdot 9}{2} = 963.050$$

$$S_M(15) = 15(95.9) + \frac{0.145 \cdot 15 \cdot 14}{2} = 1453.725$$

$$S_M(16) = 16(95.9) + \frac{0.145 \cdot 16 \cdot 15}{2} = 1551.800$$

There are 4 stints, so there are 3 pit stops:

$$\begin{aligned} T &= S_M(16) + S_M(16) + S_M(15) + S_S(10) + 3(24) \\ &= 1551.8 + 1551.8 + 1453.725 + 963.05 + 72 \\ &= 5592.375 \text{ seconds} \end{aligned}$$

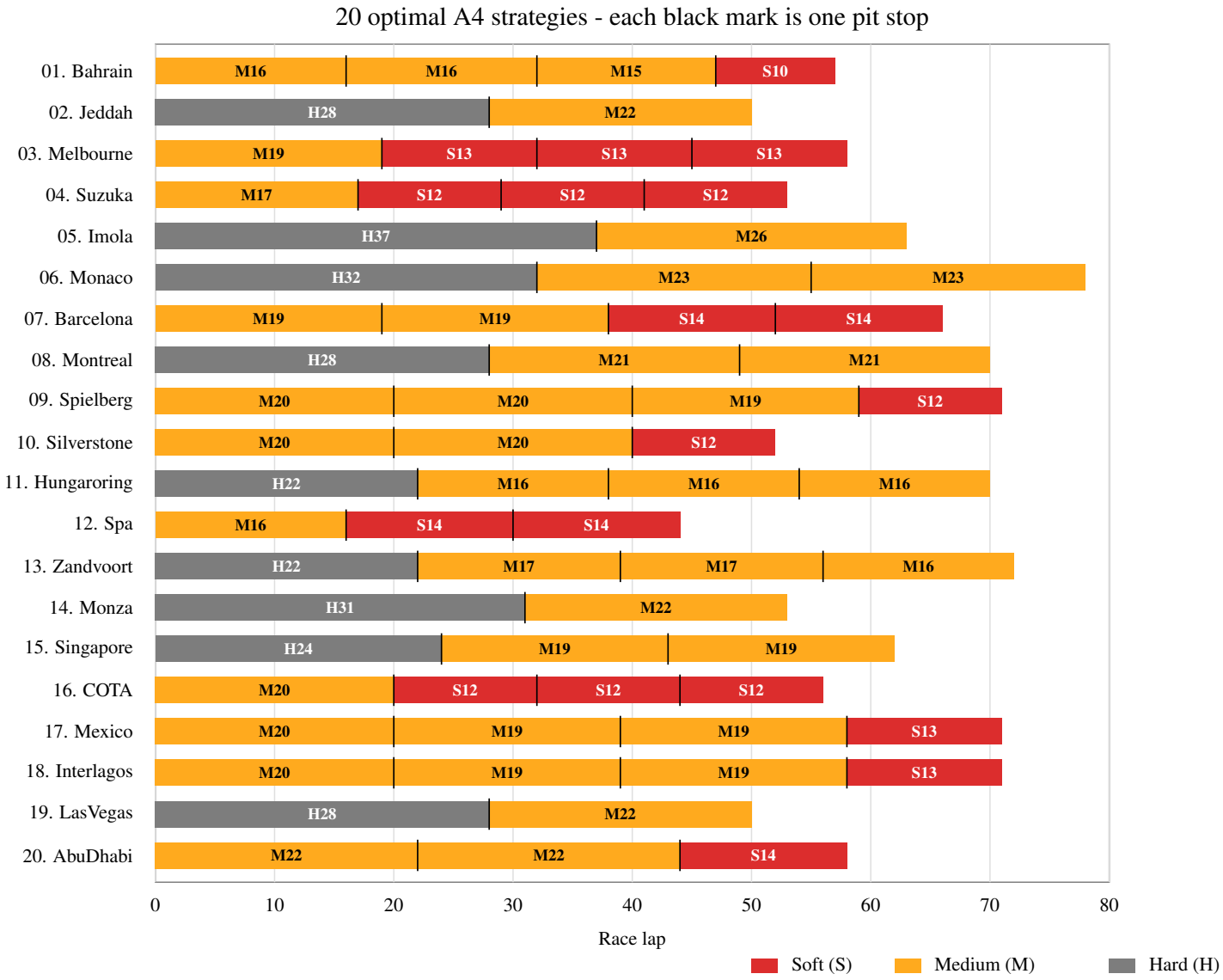


Figure 6: Visualization of all 20 strategies. The length of each segment is the number of laps in the stint; the black marks are pit stops.

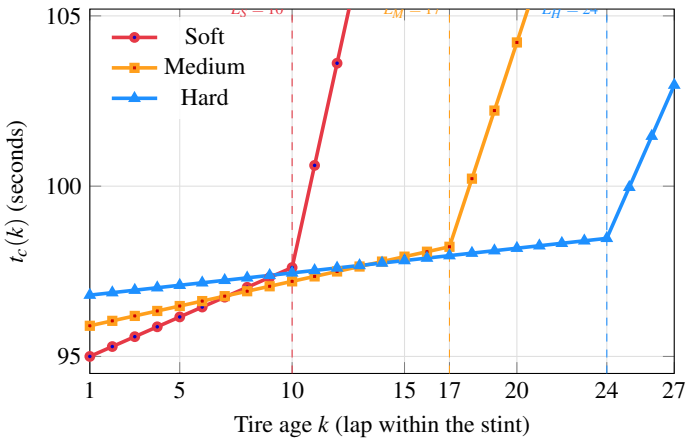


Figure 7: Each curve increases because the tire wears; after the vertical dashed line, the lap time increases very sharply.

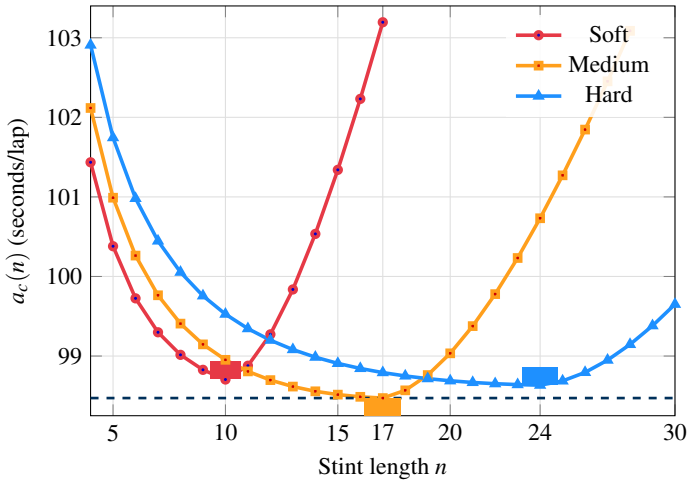


Figure 8: $M17$ has the lowest average cost; this is the standard package, not a requirement that every Medium stint must be 17 laps long.

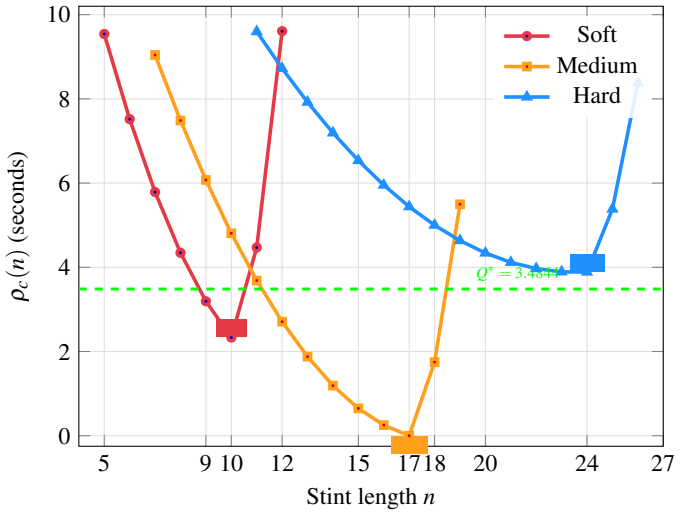


Figure 9: The minima of ρ.

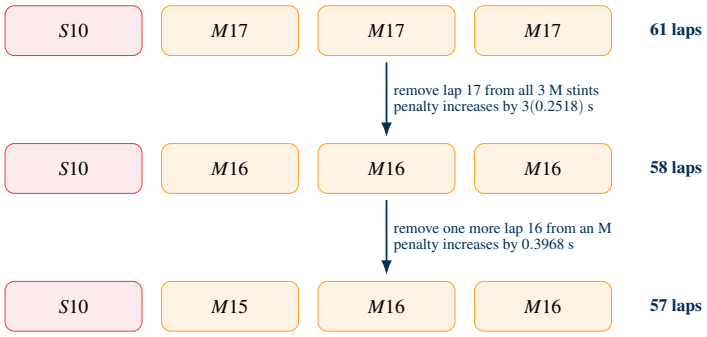


Figure 10: Adjusting around the ideal packages to match exactly 57 laps.

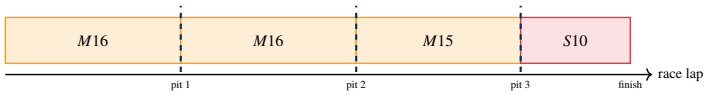


Figure 11: One execution order: pit after laps 16, 32 and 47; no pit after the final Soft stint

5.8.2 Checking Again with ρ

$$\rho_S(10) = 2.332353, \quad \rho_M(15) = 0.648529, \quad \rho_M(16) = 0.251765$$

Therefore

$$R = 2.332353 + 0.648529 + 2(0.251765) = 3.484412$$

$$T = 57\lambda - 24 + R = 5592.375$$

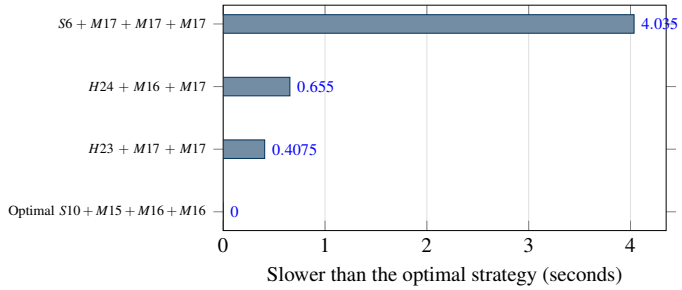


Figure 12: Comparison of some of the closest competitors. The Hard 23-lap option is only 0.4075 seconds slower, so it must be computed precisely.

6 A5. Parameter Uncertainty and Robust Strategy

According to the problem statement, A5 is split into two tasks: A5(a) builds an uncertainty model for B_c, d_c, L_c, D_c and the operational parameters; A5(b) uses that model to choose a robust strategy according to risk.

6.1 A5(a). Modeling Parameter Uncertainty

Let $\theta = (B_c, d_c, L_c, D_c)_{c\in\{S,M,H\}}$ be the parameter set of the deterministic model:

$$t_c(k) = B_c + d_c(k-1) + (D_c - d_c)(k - L_c)^+. \tag{2}$$

For a strategy consisting of m stints (c_i, n_i), the total time is computed by

$$T = \sum_{i=1}^{m} S_c(n_i) + (m-1)P, \qquad S_c(n) = \sum_{k=1}^{n} t_c(k). \tag{3}$$

Since P is also used in the computations of the deterministic model, and in F1 races tire changes depend largely on humans, the time lost to tire changes can vary by a large margin; therefore we extend the model and additionally model the uncertainty of P. We then define:

$$\theta = (B_c, d_c, L_c, D_c)_{c\in\{S,M,H\}}, P$$

In A1 to A4, we solved the problem for a fixed θ. In A5, when θ is not known with certainty, we must find a reliable strategy.

The goal of the strategy is no longer to minimize the noise-free time $T(\theta_0)$ but to choose a strategy that performs well over the whole distribution of θ

6.2 A5(a). Nominal Expected Values

The parameter values of the three compounds — Soft (S), Medium (M), Hard (H) — vary around the base values, which are the values taken from the problem statement; therefore we choose these values as the expected values of the parameters of the three compounds:

	B_c^0 (seconds)	d_c^0 (seconds/lap)	L_c^0 (laps)	D_c^0 (seconds/lap)
Soft (S)	80.0	0.20	15	3.0
Medium (M)	81.0	0.10	25	2.0
Hard (H)	82.0	0.05	35	1.5

6.3 A5(a). Noise Structure

Remark: The four tire parameters are not independently uncertain. A specific race day (track temperature, wind, grip, car condition) acts on all of them at once: a harsh day makes every compound wear faster and the cliff arrive earlier. We model this with a common noise ξ, separate from the specific noise of each compound.

Definition 6.3.1 (Common–specific noise structure)**.** The noise of each parameter is split into a shared common component and a compound-specific component:

$$\text{noise} = \underbrace{\xi}_{\substack{\text{common noise (race-day conditions)},\\ \xi\sim N(0,1)}} + \underbrace{\varepsilon_c,\ \zeta_c,\ \nu_c}_{\substack{\text{compound-specific noise},\ \sim N(0,1)\ \text{independent}}}.$$

The variable ξ is common to all three compounds: a hot day ($\xi > 0$) makes every d_c increase and every L_c come earlier simultaneously.

The influence levels of the two noises are equal.. We cannot control when the common noise (weather, car condition) acts more strongly than the specific noise (the starting position affects B_c; the engine condition affects d_c, D_c) or vice versa. There is no basis for letting either side dominate, so we choose the influence levels of the two noises to be the same. Then the influence coefficients of the common noise and the specific noise are equal, both equal to a, so the noise part of each parameter reduces to

$$a\Big(\underbrace{\xi}_{\text{common noise}} + \underbrace{\text{specific noise}}_{\varepsilon_c,\ \nu_c,\ldots}\Big).$$

Each parameter therefore needs only one influence coefficient, denoted $a_{B_c}, a_{d_c}, a_{D_c}$ (values computed in §6.5). Consequence: the common noise enters every stint with the same sign, so it does not cancel — this is the source of the tail risk that robust optimization targets.

6.4 A5(a). Probability Model for Each Parameter

Guiding principle: the distribution must match the nature of the quantity. For each parameter, we first state the physical properties to choose the form of the distribution, then write the formula together with its expectation and CoV. Determining the magnitudes of the influence coefficients is left to §6.5.

6.4.1 Base Pace B_c

Properties, and choice of distribution.. B_c is the base pace of the tire. Three properties: (i) symmetric — race-day conditions make the pace faster or slower with equal likelihood; (ii) bell-shaped — the base pace is the sum of many small independent noises (temperature, wind, measurement error), and by the central limit theorem the sum is bell-shaped; moreover, when only the expectation and variance are known, the normal is the maximum-entropy assumption (it invents the least information); (iii) proportional effect — a day that is 0.3% slower makes the 80 s tire slower by 0.24 s and the 82 s tire slower by 0.246 s, i.e. it multiplies rather than adds the same absolute amount. These three properties lead to the scaled-normal distribution (multiplicative normal).

Formula, expectation and CoV.. Applying the common–specific structure with equal influence levels:

$$B_c = B_c^0\big(1 + a_{B_c}(\xi + \nu_c)\big), \qquad \xi, \nu_c \sim N(0,1) \text{ independent.} \tag{4}$$

Since $\xi + \nu_c$ is the sum of two independent normals, B_c is still normal (Proposition 1.4.3). Expectation and CoV (proved in Section 1.5):

$$\mathbb{E}[B_c] = B_c^0 \ \text{(unbiased)}, \qquad \mathrm{CoV}(B_c) = a_{B_c}\sqrt{2}.$$

Therefore the absolute deviation $\sigma_{B_c} = B_c^0 a_{B_c}\sqrt{2}$ is proportional to the pace.

6.4.2 Degradation Coefficients d_c and D_c

Properties, and choice of distribution.. d_c (the slope before the cliff) and D_c (the slope after the cliff) are degradation rates. Three properties: (i) positive — the wear rate must be ≥ 0, and a normal distribution symmetric about 0.05 would leak probability into the negative region (the tire speeding up by itself — absurd); (ii) right-skewed — degradation can spike (harsh sun doubling it) but cannot go below 0, so the right tail is longer; (iii) proportional effect — a day that is 20% harsher makes every compound wear $\sim 20\%$ faster (Soft 0.20 shifts by ~ 0.04, Hard 0.05 shifts by ~ 0.01). These three properties lead to the log-normal distribution (since $e^x \approx 1 + x$ for small x, the multiplicative normal is exactly the linear approximation of the log-normal; we use the log-normal to keep strict positivity).

Formula, expectation and CoV..

$$d_c = d_c^0\, e^{a_{d_c}(\xi+\varepsilon_c)}, \qquad D_c = D_c^0\, e^{a_{D_c}(\xi+\varepsilon_c')}, \qquad D_c \leftarrow \max(D_c, 3d_c). \tag{5}$$

The exponent is $a_{d_c}(\xi + \varepsilon_c) \sim N(0, 2a_{d_c}^2)$, so (Proposition 1.4.2):

$$\mathbb{E}[d_c] = d_c^0\, e^{a_{d_c}^2}, \qquad \mathrm{CoV}(d_c) = \sqrt{e^{2a_{d_c}^2} - 1}$$

(and similarly for D_c). The constraint $D_c \leftarrow \max(D_c, 3d_c)$ keeps the slope after the cliff always at least three times the slope before the cliff, preserving the "cliff" nature.

6.4.3 Cliff L_c

Properties, and choice of distribution.. L_c is the cliff — the lifetime measured in laps. Properties: (i) discrete — the cliff is an integer number of laps (it cannot occur at lap 14.3), so we round a normal variable; (ii) decreases when the day is harsh — the common noise ξ enters with a minus sign so that a fast-wear day ($\xi > 0$) is also a day of an early cliff; sharing ξ with d_c builds in the physically correct negative correlation; (iii) proportional — a harsh day shortens the lifetime of every tire by the same percentage. These three properties lead to the rounded-normal distribution.

Formula.. Unlike B_c, d_c, D_c, for L_c the two noises have **different units**: the common noise is a percentage of lifetime (proportional), while the specific noise is an absolute number of laps (the measurement error of each tire set), so the two influence coefficients cannot be merged into one:

$$L_c = \mathrm{round}\big(L_c^0(1 - a_{L_c}\xi) + s_{L_c}\zeta_c\big). \tag{6}$$

Since the two components are independent, the variances add: $\sigma_{L_c} = \sqrt{(L_c^0 a_{L_c})^2 + s_{L_c}^2}$. The common part $L_c^0 a_{L_c}$ grows with the lifetime, so the more durable the tire, the harder it is to predict the cliff location in absolute laps.

6.4.4 Pit-Stop Time P (Extension)

Properties, and choice of distribution.. P is the time of one pit stop. Properties: (i) it has a physical floor — it cannot be faster than a clean stop; (ii) strongly right-skewed — it can be much slower when something goes wrong; (iii) it is a single event, not separated by compound. Heilmeier et al. (2020) and Phillips (2014) both use the log-logistic for this quantity, so we choose the log-logistic.

Formula..

$$P = P_0 + m_P\big(U/(1-U)\big)^{1/\beta}, \qquad U \sim \mathrm{Unif}(0,1). \tag{7}$$

The median is $P_0 + m_P$; the exponent $1/\beta$ controls the heaviness of the right tail (small β means a heavy tail). The values of the parameters are fixed in §6.5.

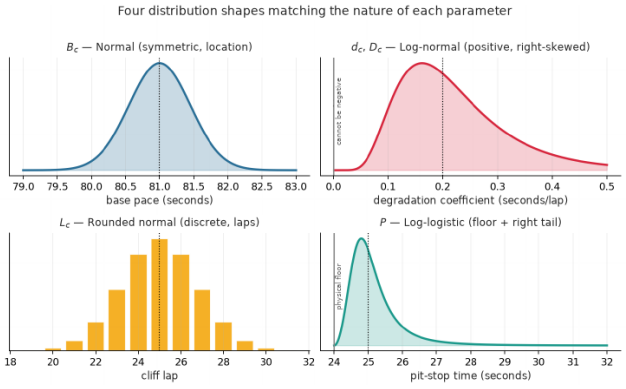


Figure 13: Four distribution shapes matching the nature of each parameter: scaled-normal (multiplicative, symmetric) for B_c; log-normal (positive, right-skewed) for d_c, D_c; rounded-normal (discrete) for L_c; log-logistic (floor + right tail) for P.

6.5 A5(a). Computing the Parameters

This part determines the magnitudes of the influence coefficients and the distribution parameters, anchored to external data.

Understanding the "priors" of Cappello & Hoegh (2025) — the foundation for every number below

Every influence coefficient computed in this part is anchored to the paper by Cappello & Hoegh (2025). That paper estimates tire parameters with Bayesian statistics, so we first need to understand a few concepts. **Prior (prior distribution).** In Bayesian statistics, an unknown parameter (for example the wear rate) is not treated as a fixed number, but as a random variable. Before looking at the data, we state our belief about it with a probability distribution, called the prior. After observing the data, the belief is updated into the posterior. The 95% credible interval of the posterior is the interval containing 95% of the posterior probability — "the parameter lies in this interval with probability 95%".

Two notations to remember. (i) $N(m, s^2)$ is the normal with expectation m and standard deviation s. The dot in $N(\cdot, 0.1^2)$ is just a placeholder for the expectation (whose value depends on the parameter), while 0.1 is the standard deviation. (ii) $N^+(m, s^2)$ is the half-normal: take the distribution $N(m, s^2)$ and cut off the negative part (keeping only values > 0); it is used for quantities that must be positive, such as the wear rate.

The actual priors in the paper (their Section 3.4):

$$\sigma_\varepsilon \sim N^+(0.3, 0.1^2), \qquad \nu \sim N^+(0.05, 0.1^2), \qquad \alpha_{\text{reset}} \sim N(69, 0.1^2).$$

Here σ_ε is the per-lap error deviation (≈ 0.3 seconds), ν is the linear wear rate (≈ 0.05 seconds/lap), and α_{reset} is the pace right after fitting new tires (≈ 69 seconds/lap — exactly the number that replaces the "·"). The paper also separates by compound: new-tire pace Hard 69.5, Medium 69, Soft 68.5 (the hard tire starts slower, the soft tire faster — physically correct).

Why the prior deviation is 0.1. The paper argues: lap times differ by only a few tenths of a second, and fluctuate by ~ 0.5 seconds per stint; therefore a prior with deviation 0.1 is "moderately tight but still reasonable". We will use exactly these benchmarks 0.1 and 0.5 to choose σ_B (§6.4.1).
Honesty — how we use them. A5 does not do Bayesian inference. We only borrow the numbers above (the prior deviations, and especially the posterior credible intervals in Tables 3–4 of the paper) as external evidence to calibrate the input uncertainty for the Monte Carlo simulation. Each coefficient below is derived step by step from a specific number; wherever there is no data, we state **[Assumption]** explicitly.

From data to influence coefficients.. The data only give us the total relative uncertainty (CoV) of each parameter. Since we fixed the influence levels of the common noise and the specific noise to be equal (§6.3), the total variance is split in half, so the influence coefficients follow directly:

$$\text{(multiplicative normal)} \quad a_{B_c} = \frac{\mathrm{CoV}}{\sqrt{2}}, \qquad \text{(log-normal)} \quad a_{d_c} = \frac{\sigma}{\sqrt{2}}, \quad \sigma = \sqrt{\ln(1+\mathrm{CoV}^2)}.$$

What remains is only to compute the CoV of each parameter from the data.

6.5.1 Influence Coefficient of B_c

The paper gives two benchmarks (the "prior" box above): the prior deviation of the pace α_{reset} is 0.1 s, and the lap time fluctuates by $\sim$ 0.5 s per stint. The day-to-day uncertainty of the base pace lies between these two benchmarks; we take $\sigma_B \approx 0.23$ s **[Source + Assumption]**. This is an absolute amount nearly the same for every tire; converting to the CoV of each base value $\mathrm{CoV}_c = 0.23/B_c^0$ and then applying $a_{B_c} = \mathrm{CoV}_c/\sqrt{2}$:

$$\boxed{\begin{aligned} B_{\mathrm{S}} &= 80\,\big(1+0.00203\,(\xi+\nu_{\mathrm{S}})\big),\\ B_{\mathrm{M}} &= 81\,\big(1+0.00201\,(\xi+\nu_{\mathrm{M}})\big),\\ B_{\mathrm{H}} &= 82\,\big(1+0.00198\,(\xi+\nu_{\mathrm{H}})\big), \end{aligned}} \tag{8}$$

giving $\sigma_{B_c} = 0.230$ s for all three compounds. The three sets differ by $< 3\%$ (the three paces differ by only $\sim 2.5\%$). The per-lap noise (driver/traffic ~ 0.3 s $= \sigma_\varepsilon$ of the paper) is strategy-neutral (added equally to every strategy), so it is not included in B_c.

6.5.2 Influence Coefficients of d_c and D_c

We extract the CoV from the posterior credible intervals in the paper instead of guessing.

Step 1 — take the real figures. Table 3 of Cappello & Hoegh reports the 95% credible intervals of the wear rate ν (which is exactly our d_c):

$$\text{Hard: } \hat{\nu} = 0.054, \quad \mathrm{CI}_{95\%} = [0.004, 0.133],$$
$$\text{Medium: } \hat{\nu} = 0.060, \quad \mathrm{CI}_{95\%} = [0.009, 0.120].$$

Step 2 — convert the credible interval into a standard deviation. When the posterior is approximately normal, a 95% credible interval has the form $\hat{\nu} \pm 1.96\,\sigma$. Why 1.96: for the standard normal variable $N(0,1)$, the 97.5% quantile equals 1.96 — meaning 2.5% probability is left in each tail, leaving exactly 95% in the middle. Therefore the width of the interval is (upper bound) $-$ (lower bound) $= 2 \times 1.96\,\sigma = 3.92\,\sigma$, hence

$$\sigma = \frac{\text{upper bound} - \text{lower bound}}{3.92}.$$

Applying this: Hard tire $\sigma = \dfrac{0.133 - 0.004}{3.92} \approx 0.0329$; Medium tire $\sigma = \dfrac{0.120 - 0.009}{3.92} \approx 0.0283$.

Step 3 — convert to CoV (relative uncertainty). Divide σ by the central value itself:

$$\mathrm{CoV}_{\text{Hard}} = \frac{0.0329}{0.054} \approx 0.61, \qquad \mathrm{CoV}_{\text{Med}} = \frac{0.0283}{0.060} \approx 0.47.$$

These two numbers bracket 0.50 from both sides. So the target $\mathrm{CoV}_d \approx 0.50$ is not an arbitrary assumption but a number extracted from the data (the average of the two compounds). **[Source + computation]**.

Step 4 — convert CoV to the log-SD σ. Proposition 1.4.2 gives $\mathrm{CoV} = \sqrt{e^{\sigma^2} - 1}$ for the log-normal. Solving the equation with $\mathrm{CoV} = 0.5$:

$$\sqrt{e^{\sigma^2} - 1} = 0.5 \;\Rightarrow\; e^{\sigma^2} = 1.25 \;\Rightarrow\; \sigma^2 = \ln 1.25 \approx 0.2231 \;\Rightarrow\; \sigma \approx 0.472.$$

Table 3 gives different CoVs by compound — Medium ≈ 0.49, Hard ≈ 0.60 (the slower a tire wears, the weaker the signal and so the noisier it is in relative terms). The Soft tire has no separate CI; following that trend (Soft wears fastest) we take a lower CoV, ≈ 0.42 **[Assumption based on the trend]**. Applying $a_{d_c} = \sigma_c/\sqrt{2}$:

$$\boxed{\begin{aligned} d_{\mathrm{S}} &= 0.20\,e^{0.285\,(\xi+\varepsilon_{\mathrm{S}})}, \quad (\mathrm{CoV} \approx 0.42),\\ d_{\mathrm{M}} &= 0.10\,e^{0.328\,(\xi+\varepsilon_{\mathrm{M}})}, \quad (\mathrm{CoV} \approx 0.49),\\ d_{\mathrm{H}} &= 0.05\,e^{0.392\,(\xi+\varepsilon_{\mathrm{H}})}, \quad (\mathrm{CoV} \approx 0.60). \end{aligned}} \tag{9}$$

The absolute deviation σ_{d_c} is Soft 0.084, Medium 0.049, Hard 0.030 s/lap — the Soft tire has the largest absolute variability (physically correct: it wears fastest). For D_c, anchoring to the degradation acceleration β in Table 4 gives CoV ≈ 0.40 (Medium, Hard); we give the hard tire slightly more and the soft tire slightly less **[Assumption]**:

$$\boxed{\begin{aligned} D_{\mathrm{S}} &= 3.0\,e^{0.247\,(\xi+\varepsilon'_{\mathrm{S}})}, \quad (\mathrm{CoV} \approx 0.36),\\ D_{\mathrm{M}} &= 2.0\,e^{0.272\,(\xi+\varepsilon'_{\mathrm{M}})}, \quad (\mathrm{CoV} \approx 0.40),\\ D_{\mathrm{H}} &= 1.5\,e^{0.304\,(\xi+\varepsilon'_{\mathrm{H}})}, \quad (\mathrm{CoV} \approx 0.45), \end{aligned}} \qquad D_c \leftarrow \max(D_c, 3d_c). \tag{10}$$

6.5.3 Influence Coefficient of L_c

The common influence coefficient $a_{L_c} = 0.10$ **[Assumption]**: a harsh day cuts $\sim 10\%$ of the lifetime (since it is proportional, the 35-lap Hard tire loses 3.5 laps, while the 15-lap Soft tire loses only 1.5 laps). The specific noise is an absolute fluctuation $s_{L_c} = 1.8$ laps **[Assumption]** (the measurement error of each tire set). Combined, $\sigma_{L_c} = \sqrt{(L_c^0 \cdot 0.10)^2 + 1.8^2}$:

$$\boxed{\begin{aligned} L_{\mathrm{S}} &= \mathrm{round}\big(15(1-0.10\,\xi) + 1.8\,\zeta_{\mathrm{S}}\big), \quad \sigma_{L_{\mathrm{S}}} \approx 2.34,\\ L_{\mathrm{M}} &= \mathrm{round}\big(25(1-0.10\,\xi) + 1.8\,\zeta_{\mathrm{M}}\big), \quad \sigma_{L_{\mathrm{M}}} \approx 3.08,\\ L_{\mathrm{H}} &= \mathrm{round}\big(35(1-0.10\,\xi) + 1.8\,\zeta_{\mathrm{H}}\big), \quad \sigma_{L_{\mathrm{H}}} \approx 3.94. \end{aligned}} \tag{11}$$

[Assumption]: the cliff timing is almost unobservable in real data (teams always pit before it), so the coefficients here have a physical basis rather than being measurements; the model form (multiplicative, signed, rounded) is rigorous.

6.5.4 Parameters of P

Take P_0=24, m_P=1, β=3: floor 24 s, median $P_0 + m_P = 25$ s (**[Source + computation]**: chosen so that the median matches the nominal value $P = 25$ of the problem), β=3 **[Assumption]** for a right tail up to $\sim$ 30+ seconds. Illustrating the tail with quantiles: Illustrating the tail with quantiles (very intuitive). Substituting each level U into $P = 24 + (U/(1-U))^{1/3}$ tells us "how bad a pit day can be":

U	0.50	0.75	0.90	0.95	0.99
P (seconds)	25.00	25.44	26.08	26.67	28.63

Reading the table: the median (50%) is exactly 25 s (matching the nominal value of the problem); 10% of pit stops are slower than 26.08 s; and the worst 1% exceed 28.6 s — exactly the shape of "usually fast, occasionally broken".

Each pit adds $\bar{P} \approx 25$ s to the expectation $\mathbb{E}[T]$ — this is the main effect of the number of pit stops. The pit variance is small ($(m-1)\sigma_P^2$ with $\sigma_P \approx 0.98$; all 3 pits add only ~ 1.7 s of deviation), so the pit is not the main source of σ; the dominant source of instability is the cliff exposure through the common noise ξ (§6.8).

6.6 A5(a). Propagating Uncertainty by Monte Carlo

Having the input distributions (4)–(7), we need the output distribution of T. But $T = f(B,d,L,D,P)$ is nonlinear and rugged: it has a cliff (a step function), the rounding of L_c, and a max. Hence there is no closed formula for the distribution of T; the analytic approximation (delta-method, Theorem 1.6.1) is only locally linear and breaks down exactly at the cliff — the most important place.

Monte Carlo algorithm.. For each fixed strategy:

1. draw $\xi \sim N(0,1)$ and then $\varepsilon_c, \zeta_c, \nu_c$, generate a set (B_c, d_c, L_c, D_c) and one P;
2. compute the deterministic T by (3);
3. repeat $n_{\mathrm{mc}} = 90{,}000$ times $\rightarrow$ obtain the empirical distribution of T.

By the law of large numbers, the empirical distribution converges to the true distribution with error $\sim 1/\sqrt{n_{\mathrm{mc}}} \approx 0.33\%$. Role: Monte Carlo here **does not search for** strategies — it **scores the robustness** of the strategies derived in A1–A4. (The same strategy, 90,000 different "race days".) Pit

stops are drawn independently for each stop ($T = \sum_i S_c(n_i) + \sum_j P_j$, each P_j a separate sample) — more physically correct than "one common P" and giving a pit variance linear in the number of stops.

The confidence band widens with the variable — all three compounds.. Theorem 1.6.1 gives a complementary analytic form: the 95% band of the lap time widens linearly with tire age k (due to the uncertain slope d_c) and then explodes at the cliff (due to the uncertain location L_c). Figure 14 draws this for all three compounds: the band widens faster for the Soft tire (the CoV of d is low but the absolute σ_d is the largest) and the explosion point moves further out (L_S=15 → L_M=25 → L_H=35) while becoming wider (σ_L increases 2.34 → 3.08 → 3.94 laps according to (11)). This is the intuitive quantitative basis for Theorem 1.7.1.

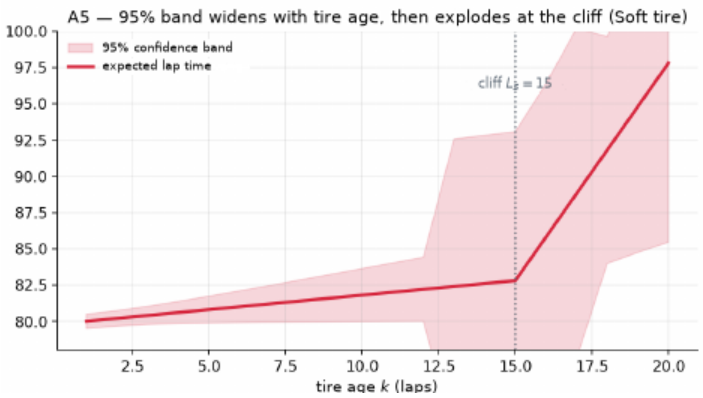


Figure 14: The 95% band of the lap time for the three compounds (Theorem 1.6.1): narrow when the tire is new, widening with tire age, exploding at the cliff L_S=15, L_M=25, L_H=35. The more durable the tire, the later the cliff and the harder it is to predict (σ_L grows).

6.7 A5(b). Risk Evaluation Criteria

Definition 6.7.1 (Value at risk VaR and CVaR)**.** Given a confidence level $\alpha \in (0,1)$ (we use $\alpha = 0.90$).

- The value at risk $\mathrm{VaR}_\alpha[T]$ is the α-level quantile of T: the smallest number v such that $\Pr(T \le v) = \alpha$. Interpretation: with probability $\alpha\,(= 90\%)$, the race time does not exceed VaR_α — it is the "bad day" benchmark. The subscript "90" in VaR_{90} is exactly $\alpha = 0.90$ written as a percentage. The complement $1-\alpha = 0.10$ is the significance level (the proportion of days worse than the benchmark).
- The conditional value at risk $\mathrm{CVaR}_\alpha[T] = \mathbb{E}[T \mid T \ge \mathrm{VaR}_\alpha[T]]$ is the mean of T over the $(1-\alpha) = 10\%$ worst cases (the right tail).

Why these are two new evaluation variables: unlike the expectation $\mathbb{E}[T]$ (which measures the average) and the standard deviation σ (which measures symmetric dispersion), VaR and CVaR specifically measure tail risk — VaR_{90} gives the bad-day threshold, while CVaR_{90} tells how bad those bad days are (the average of the 10% worst scenarios). In racing, it is exactly this tail that decides.

Each strategy gives a distribution of T; we summarize it by three numbers (Definition 6.7.1): the expectation $\mathbb{E}[T]$ (average performance), the standard deviation σ (stability — recall that the 95% band has width 3.92σ, so a small σ means "consistently good over the middle 95%"), and CVaR_{90} (tail risk). Why all three are needed, not just the expectation: races are lost in the tail — a strategy that is usually fine but occasionally catastrophic is a poor strategy; σ and CVaR capture exactly that tail. In §6.8, for each criterion we will derive the strategy that minimizes it, and then cross-evaluate all three.

6.8 A5(b). Building the Set of Candidate Strategies

Method (why not exhaustive search).. The strategy space is every way of dividing N=50 laps into stints and assigning compounds — too large for exhaustive search, and A5 does not aim to find the absolute optimum. Instead we do something more rigorous in terms of reasoning: for each risk criterion (the expectation $\mathbb{E}[T]$, the standard deviation σ, CVaR_{90}), we derive mathematically the structural features that a strategy must have for that criterion to be smallest. Each feature yields one candidate strategy; we add a few fixed strategies as reference benchmarks; finally we cross-evaluate the whole group on all three criteria (Table 4). It may be that no candidate is optimal in every respect — but the strategies derived are well-founded, not invented.

A shared structural lemma: stints are bounded by the cliff.. Before considering each criterion, we establish a fact that simplifies all the reasoning that follows.

Proposition 6.8.1 (Degradation saving from adding a pit, and the cliff-bounded consequence)**.** On a compound with wear rate d (ignoring the cliff), splitting a stint of length $n = n_1 + n_2$ into two stints n_1, n_2 (fitting new tires in between) reduces the total accumulated degradation time by exactly $d\,n_1 n_2$. Therefore adding a pit is only "worth it" in terms of degradation when the saving exceeds the pit cost $\bar{P}$; with a balanced split $n_1 = n_2 = n/2$, the condition is $n > n_c^\star := 2\sqrt{\bar{P}/\bar{d}_c}$.

Proof. The degradation part of the stint cost is $d \cdot \frac{n(n-1)}{2}$ (the arithmetic series $0, d, 2d, \ldots$). With $n = n_1 + n_2$, expand $n(n-1) = (n_1+n_2)(n_1+n_2-1) = n_1(n_1-1) + n_2(n_2-1) + 2n_1n_2$. Hence

$$\underbrace{d\tfrac{n(n-1)}{2}}_{\text{one long stint}} - \underbrace{\left(d\tfrac{n_1(n_1-1)}{2} + d\tfrac{n_2(n_2-1)}{2}\right)}_{\text{two stints, new tires in between}} = d\,n_1 n_2 > 0,$$

exactly as stated (new tires "reset the wear clock"). The balanced split gives the largest saving $\approx \bar{d}_c (n/2)^2$; requiring it to exceed $\bar{P}$ immediately gives the threshold $n_c^\star$. □

Consequence (key). Using the expected values (the log-normal gives $\bar{d}_c = d_c^0 e^{\sigma_d^2/2} = 1.112\,d_c^0$, i.e. 11% higher than nominal — uncertainty increases the degradation cost) and $\bar{P} \approx 25.2$ s:

$$n_{\mathrm{S}}^\star \approx 2\sqrt{25.2/0.222} \approx 21.3, \quad n_{\mathrm{M}}^\star \approx 30.1, \quad n_{\mathrm{H}}^\star \approx 42.6 \text{ (laps)}.$$

But every nominal cliff is smaller than this threshold: L_S=15 < 21.3, L_M=25 < 30.1, L_H=35 < 42.6. That is, before a stint is long enough to be "worth an extra pit because of degradation", it has already reached the cliff. **So the constraint that decides the stint length is the cliff, not the degradation–pit trade-off:** $n_i \lesssim L_{c_i}$ (minus a safety margin according to Theorem 1.7.1). From this, the number of pit stops follows directly from the cliff — number of stints $\approx \lceil N/(L_c - \text{margin}) \rceil$:

$$\begin{aligned} \text{all Soft} &\to \lceil 50/13 \rceil = 4 \text{ stints (3 pits)};\\ \text{all Medium} &\to \lceil 50/23 \rceil = 3 \text{ stints (2 pits)};\\ \text{all Hard} &\to 2 \text{ stints (1 pit)}. \end{aligned}$$

(a) Minimizing $\mathbb{E}[T]$ — deriving the formula from scratch, then comparing by number of pit stops.. Start from the basic decomposition: the total time equals the total time of the stints plus the total time of the pit stops,

$$T = \sum_{i=1}^{m} S_c(n_i) + \sum_{j=1}^{m-1} P_j.$$

Taking the expectation of both sides, using the linearity of expectation:

$$\mathbb{E}[T] = \sum_{i=1}^{m} \mathbb{E}[S_c(n_i)] + (m-1)\bar{P}.$$

Expand one stint using the definition $S_c(n) = \sum_{k=1}^{n} t_c(k)$ and add term by term:

$$\begin{aligned} S_c(n) &= \sum_{k=1}^{n} \left[B_c + d_c(k-1) + (D_c - d_c)(k - L_c)^+\right]\\ &= nB_c + d_c\frac{n(n-1)}{2} + (D_c - d_c)\Phi_c(n). \end{aligned}$$

Taking the expectation of a stint (the log-normal gives $\bar{d}_c = d_c^0 e^{\sigma_{d,c}^2/2}$, $\bar{\varphi}_c(n) = \mathbb{E}[\Phi_c(n)]$), and summing, we obtain the final four-term formula:

$$\begin{aligned} \mathbb{E}[T] = &\sum_i n_i B_{c_i}^0 + \sum_i \bar{d}_{c_i}\frac{n_i(n_i-1)}{2}\\ &+ \sum_i (\bar{D}_{c_i} - \bar{d}_{c_i})\bar{\varphi}_{c_i}(n_i) + (m-1)\bar{P}. \end{aligned} \tag{12}$$

Why we must compare by number of pit stops. Soft has the fastest base (80), but all-Soft needs 3 pits (the cliff-bounded lemma). That is not necessarily the minimum: perhaps replacing some Soft with Medium/Hard to need only 2 pits or 1 pit would be smaller thanks to saving 25–50 s of pit time? Only computing it tells us. For each pit-count class, we choose the best representative in the class and then compute (12) (Table 3).

Table 3: Decomposition of $\mathbb{E}[T]$ (12) for the best representative of each pit-count class (seconds). "base+pit" is shown separately to reveal the speed–number-of-stops trade-off.

Representative	base	degradation	pit	cliff	$\mathbb{E}[T]$	base+pit
3 pits All-Soft S13·S13·S12·S12	4000.0	62.5	75.6	3.6	**4141.7**	4075.6
2 pits S–M–M S15·M18·M17	4035.0	54.9	50.4	5.6	4146.0	4085.4
1 pit M25·M25	4050.0	66.7	25.2	12.4	4154.3	4075.2

Why the representative of each class is chosen this way (proof of optimality within the class): 3 pits — to have the smallest base we run Soft everywhere (80 < 81 < 82), and all-Soft by itself needs 4 stints, so it is the fastest 3-pit

representative. 2 pits — 3 stints cover 50 laps; maximizing the number of Soft laps is bounded ($\le$ 15/stint), but placing 2 Soft stints exactly at the cliff (as in S15·S15·M20) incurs a heavy cliff penalty, so a good representative is 1 Soft + 2 Medium with wide margins (S–M–M). 1 pit — 2 stints cover 50, Soft cannot cover them ($\le$ 30), so a late-cliff tire is forced; M25·M25 is the fastest feasible option.

Reading the table to conclude. The **base+pit** column is nearly equal (4075–4085): dropping one pit saves $\bar{P} \approx 25$ s but forces ~ 25 laps to switch from Soft (80) to Medium (81), i.e. +25 s — almost cancelling out. So the speed–number-of-stops trade-off is shallow; what separates them is the two remaining columns. The short stints of All-Soft keep both degradation (62.5, even though Soft wears fast) and the cliff penalty (3.6, thanks to the margin) low; whereas M25·M25 has long stints exactly at the cliff, so the cliff penalty jumps to 12.4 and degradation to 66.7. **Conclusion: All-Soft (3 pits) minimizes $\mathbb{E}[T] = 4142.8$ s** (MC, Table 4) — the extra pit is compensated by the fast base, and short stints win on degradation + cliff.

(b) Minimizing σ — stating the formula and then minimizing it like an inequality problem.. (Recall: a small $\sigma \Leftrightarrow$ the central 95% band 3.92σ is the narrowest, i.e. "consistently good over the middle 95%".) Formula. Decompose the variance by the law of total variance around the common factor ξ:

$$\begin{aligned} \mathrm{Var}[T] &= \mathrm{Var}_\xi\big(\mathbb{E}[T \mid \xi]\big) + \mathbb{E}_\xi\big(\mathrm{Var}[T \mid \xi]\big) + (m-1)\sigma_P^2, \\ g &= \frac{\partial T}{\partial \xi} = \sum_i \frac{\partial S_c(n_i)}{\partial \xi}. \end{aligned} \tag{13}$$

Since a harsh day slows every stint with the same sign, the derivatives $\partial S_c/\partial\xi$ add in the same direction (they do not cancel) — so g is a sum of terms with the same sign, and the largest term is the cliff exposure: a stint close to the cliff has $\partial S_c/\partial\xi$ containing a very large $(D_c - d_c)\sigma_{L,c}$. Collecting this part, the contribution of the cliff to the variance is

$$\begin{aligned} V_{\text{cliff}} &= \sum_{i=1}^{m} w_i\,(D_{c_i} - d_{c_i})^2\,\sigma_{L,c_i}^2, \\ w_i &= \Pr(\text{stint } i \text{ reaches the cliff}) \in [0,1]. \end{aligned} \tag{14}$$

Minimizing like an inequality problem. (14) is a sum of nonnegative terms. Expanding and simplifying: each term is ≥ 0, with equality when w_i=0 (a stint with a margin, not reaching the cliff); and for a stint that must be close to the cliff, the term is bounded below by the $(D_c - d_c)^2\sigma_{L,c}^2$ of that compound itself. Therefore

$$V_{\text{cliff}} \ge \big(\#\text{stints near the cliff}\big)\,\min_c (D_c - d_c)^2\sigma_{L,c}^2,$$

and the minimum is attained when: (i) there are as few stints near the cliff as possible, (ii) the stints that must be near the cliff use a tire with small $(D-d)$ and σ_L. Consider the feasible cases (covering 50 laps, still using Soft speed) and compute (14) for each:

Nominal S15·S15·M20	:	2 Soft *exactly* at the cliff (w=.5) $\Rightarrow V_{\text{cliff}} \approx 50.6$,
Conservative M25·M25	:	2 Medium *exactly* at the cliff $\Rightarrow$ 39.5 (plus long-stint degradation),
Margin S13·S13·M24	:	3 partially exposed $\Rightarrow$ 33.8,
All-Soft S13·S13·S12·S12	:	4 Soft lightly exposed ($w \approx .1$–.2) $\Rightarrow$ 28.7,
S–M–M S15·M18·M17	:	**1** Soft near the cliff; 2 Medium with wide margins ($w \approx 0$) $\Rightarrow$ **24.9**.

S–M–M attains the minimum: exactly one stint near the cliff (the Soft one, unavoidable if Soft speed is wanted), the rest with wide margins. Computing the total by MC: σ in the order S–M–M 40.0 < All-Soft 45.2 < Margin 49.8 < Nominal 57.1 < Conservative 70.5 — **S–M–M minimizes** σ. (Conservative is highest because 25-lap stints add extra accumulated degradation variance.)

Note — why the pit is not the culprit. The pit term $(m-1)\sigma_P^2$ in (13) is very small: $\sigma_P^2 \approx 0.96$, so all 3 pits add only ≈ 1.7 s of deviation — negligible. The whole σ is dominated by the cliff exposure through the common ξ (14), and S–M–M wins because it minimizes that — not because of the number of pit stops.

(c) Minimizing CVaR_{90} — the same inequality, with a heavier tail.. Formula. Using the Rockafellar–Uryasev form (or the tail identity):

$$\mathrm{CVaR}_\alpha[T] = \mathrm{VaR}_\alpha[T] + \frac{1}{1-\alpha}\,\mathbb{E}\big[(T - \mathrm{VaR}_\alpha[T])^+\big], \qquad \alpha = 0.90. \tag{15}$$

The 10% worst days are harsh days (large ξ): expanding $T \approx T_0 + g\,\xi +$ (cliff bite), where the "cliff bite" is convex (when ξ pulls the cliff earlier, the penalty grows superlinearly). Substituting into (15), the tail part $\frac{1}{1-\alpha}\mathbb{E}[(T - \mathrm{VaR})^+]$ is a sum of nonnegative convex contributions from exactly the stints exposed to the cliff — the same structure as (14) but with even larger tail weights. Repeating exactly the inequality argument in (b): the minimum is attained with as few stints near the cliff as possible + the largest margin (so that the bite does not occur even on a harsh day). MC computation: CVaR_{90} in the order S–M–M 4233.8 < All-Soft 4237.7 < Margin 4257.6 < Nominal 4278.6 < Conservative 4318.6 — **S–M–M minimizes** CVaR, exactly the solution of (b). The two risk criteria converge to one structure because both are dominated by the nonnegative total cliff exposure.

6.8.1 The Set of Five Candidates

Five candidates, each tied to one structural argument above (or serving as a reference benchmark): (1) Nominal balance S15·S15·M20 — the deterministic A1–A4 solution (no margin), the benchmark "if uncertainty is ignored"; (2) All-Soft 4 stints — the $\mathbb{E}[T]$-minimizing solution in (a); (3) S–M–M S15·M18·M17 — the σ- and CVaR-minimizing solution in (b),(c); (4) Safety Margin S13·S13·M24 — candidate (1) plus a margin (applying Theorem 1.7.1); (5) Conservative M25·M25 — the fewest pits (1 pit), the benchmark "fewest pits regardless of cliff exposure".

6.9 A5(b). Robust Strategy and Risk Analysis

Table 4: Risk profile of the candidate strategies (90,000 samples, independent pits, log-normal model + log-logistic P). The best value in each column is in bold.

Strategy	$\mathbb{E}[T]$ (seconds)	σ (seconds)	CVaR_{90} (seconds)
Nominal optimum S15·S15·M20	4151.1	57.1	4278.6
Safety margin S13·S13·M24	4147.4	49.8	4257.6
S–M–M S15·M18·M17	4147.3	**40.0**	**4233.8**
All-Soft 4 stints S13·S13·S12·S12	**4142.7**	45.2	4237.7
Conservative M25·M25	4158.9	70.5	4318.6

Cross-evaluation: reading the results table along the Pareto front

No candidate wins all three columns — this is a genuine Pareto trade-off, exactly as predicted: each strategy is the minimizer of one criterion. **(1) $\mathbb{E}[T]$:** All-Soft 4 stints is the fastest on average (4142.7) — matching reasoning (a) (the Soft base advantage outweighs the extra pit cost). But it ranks only 4th in σ and 2nd in CVaR: it pays in the tail because of 3 pits + 4 cliff exposures. **(2) σ and CVaR:** S–M–M wins both (40.0 and 4233.8), exactly as in reasoning (b),(c), while its $\mathbb{E}[T]$ is only 4.6 s (0.11%) slower than All-Soft — trading 0.11% of expectation for a $\sim 11\%$ reduction in σ and the lowest tail. This is the recommended robust choice. **(3) Margin strictly dominates Nominal:** S13·S13·M24 is better than S15·S15·M20 in all three columns — confirming Theorem 1.7.1: backing off 1–2 laps before the cliff is never a loss, so the "nominal balance" is eliminated (dominated). **(4) Fewer pits $\neq$ robust:** Conservative M25·M25 has only 1 pit but is the worst in σ and CVaR — because both stints stop exactly at the cliff L_M=25, with maximal cliff exposure. This refutes the intuition "fewer pits is always safer".

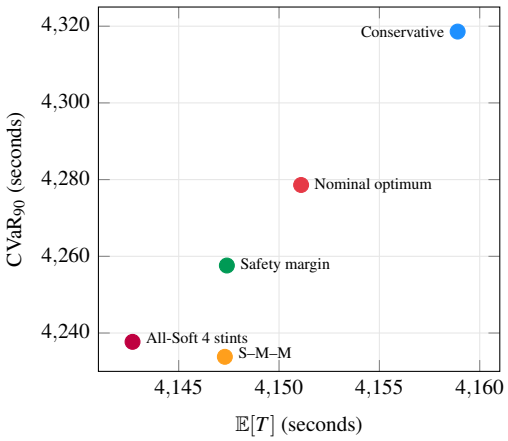


Figure 15: Expectation–tail-risk frontier: the horizontal axis is $\mathbb{E}[T]$, the vertical axis is CVaR_{90}. S–M–M lies in the lower-left corner (most robust: lowest tail, near-optimal expectation); Conservative is in the upper-right corner (worst on both). All-Soft is furthest left in $\mathbb{E}[T]$ but slightly higher than S–M–M in CVaR.

6.10 Scope: What Is Proved and What Is a Choice

Derived mathematically (rigorous): (i) stints are bounded by the cliff rather than by the degradation–pit trade-off (Proposition 6.8.1, with proof); (ii) the number of pits for each compound follows from the cliff; (iii) the cliff term dominates σ and is eliminated by the margin (Theorem 1.6.1); (iv) the CVaR tail = harsh days due to the common ξ (Definition 6.3.1); (v) the margin dominates the nominal solution (Theorem 1.7.1). **A choice, not a theorem (and why it is acceptable):** (1) we evaluate a finite set of candidates, not an exhaustive search — acceptable because the cliff-bounded lemma has already narrowed the reasonable space to a few families, and each candidate is the structural minimizer of one criterion so together they

bracket the Pareto front; we may miss a length that is slightly longer but this does not change the picture. (2) The size of the margin (1–2 laps) and the exact number of pits are fixed by Monte Carlo, not by a closed formula — because $\bar{\varphi}$ and the common-ξ tail have no compact closed form; we derive the direction (shorter than $n^\star$, counted from the cliff) mathematically and let MC refine the integers. (3) Choosing α=0.90 for CVaR is a matter of risk appetite and cannot be derived — we state this clearly; switching to 0.95 only reinforces the same ranking (a heavier tail penalizes the cliff-exposed solutions even more).

6.11 Conclusion

Three core points:

1. The uncertainty model matches the physical nature of each parameter (B_c multiplicative normal; d_c, D_c log-normal; L_c rounded normal; P log-logistic), linked through the common race-day factor ξ to build in the correct correlation; every coefficient is derived from the posterior credible intervals of Cappello & Hoegh (2025), not assigned arbitrarily.
2. The main source of instability is cliff exposure through the common ξ, not the pit variance (which is negligible).
3. The recommended robust strategy is **S–M–M** ($S15 \cdot M18 \cdot M17$): it minimizes both σ and CVaR_{90}, while its expectation is only $\approx 0.11\%$ slower than the fastest solution (All-Soft).

References.. C. Cappello, A. Hoegh (2025), A State-Space Approach to Modeling Tire Degradation in Formula 1 Racing, arXiv:2512.00640 — half-normal for the degradation rate, observation error ~ 0.3 seconds, skew-t for asymmetry, fuel 110 kg$\rightarrow$ 0, degradation increasing over time. A. Heilmeier, M. Graf, J. Betz, M. Lienkamp (2020), Application of Monte Carlo Methods to Consider Probabilistic Effects in a Race Simulation for Circuit Motorsport, Applied Sciences 10(12):4229, DOI:10.3390/app10124229 — log-logistic pit time, Monte Carlo simulation. T. Phillips (2014) — log-logistic pit model.

7 A6. Calibrating the Model from Real Data

7.1 A6(a). Choice of Races, Car and Data Extraction

Additional assumptions used in the report.

- Linear fuel effect. The amount of fuel decreases almost linearly with the number of laps (burn rate $\approx$ 1.6–1.8 kg/lap), and each kg of fuel makes a lap slower by about 0.03s.
- The starting lap (standing start), the pit-entry lap (in-lap), the pit-exit lap (out-lap) and the laps run under yellow flags or the Safety Car are excluded from every regression, because they do not reflect stable tire performance.
- One compound per stint. Each stint uses exactly one compound, exactly as defined in the problem statement.

A single race is a weak basis for calibrating the model, because the other effects act in a way specific to that race. Using several races with different stint structures opens additional independent perspectives, helping to verify the conclusions. Therefore the research team analyzes three races of the same driver, Max Verstappen, in the 2023 season, with the following criteria:

- **Keeping the driving style fixed.** Using the same driver and the same car removes the variation due to skill and tire management between drivers, isolating tire performance better.
- **Undisturbed air.** Verstappen won all three races from the front of the grid, so his data lie in clean air, little disturbed by following other cars or by congestion when overtaking.
- **All dry, all three stints, a variety of compounds.** All three races were dry and all had two pit stops (three stints). The set of three strategies covers all three compounds and, crucially, gives each compound three stints at different times (fuel levels), enabling both estimation and verification.
- **A variety of track characteristics.** The three tracks have different profiles, together with different characteristics.
- **Two information-rich structures for verification.** Bahrain has two stints on the soft tire and Hungary has two stints on the medium tire. These same-compound pairs are a direct test of the fuel model. Moreover, Hungary ran Medium $\rightarrow$ Hard $\rightarrow$ Medium, "sandwiching" the Hard stint between two Medium stints, which is an ideal structure for comparing Medium with Hard.

Table 5: Summary of the 3 races

Race (track)	Characteristics	N	Pit laps	Stint lengths	Strategy
Bahrain (Sakhir)	hot, abrasive	57	14, 36	14 / 22 / 21	Soft $\rightarrow$ Soft $\rightarrow$ Hard
Spain (Barcelona)	fast corners, high wear	66	26, 52	26 / 26 / 14	Medium $\rightarrow$ Hard $\rightarrow$ Soft
Hungary (Hungaroring)	slow corners, hot	70	23, 51	23 / 28 / 19	Medium $\rightarrow$ Hard $\rightarrow$ Medium

Data processing

For each race, before the regression the research team removes the following laps:

- Lap 1 (standing start, slow due to accelerating from the start line), the pit-entry laps (in-laps) and pit-exit laps (out-laps, several seconds slower because of passing through the pit lane), together with outliers due to traffic or yellow flags.

Data extraction

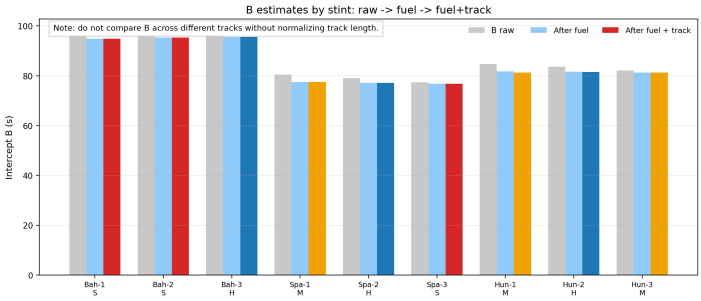


Figure 16: Estimates of B by stint before and after correcting for fuel and track.

The code below extracts the lap times of each stint by compound for Verstappen in all three Grands Prix, using the official FastF1 library.

```
import fastf1
import pandas as pd

fastf1.Cache.enable_cache("./fastf1_cache")

YEAR = 2023
DRIVER = "VER"
GPS = ["Bahrain", "Spain", "Hungary"]


def extract_one(gp):
    ses = fastf1.get_session(YEAR, gp, "R")
    ses.load(telemetry=False, weather=True)

    laps = ses.laps.pick_drivers(DRIVER).copy()
    laps["LapTimeS"] = laps["LapTime"].dt.total_seconds()

    cols = [
        "LapNumber",
        "LapTimeS",
        "Stint",
        "Compound",
        "TyreLife",
        "PitInTime",
        "PitOutTime",
        "TrackStatus"
    ]

    laps = laps[cols].reset_index(drop=True)

    laps["is_outlap"] = laps["PitOutTime"].notna()
    laps["is_inlap"] = laps["PitInTime"].notna()
    laps["is_lap1"] = laps["LapNumber"] == 1

    laps["clean"] = ~(
        laps["is_lap1"] | laps["is_inlap"] | laps["is_outlap"]
    ) & (laps["TrackStatus"].astype(str) == "1")

    print(f"\n=== {DRIVER} - {YEAR} {gp} GP ===")

    for st, grp in laps.groupby("Stint"):
        comp = grp["Compound"].mode().iat[0]
        c = grp[grp["clean"]]

        print(
            f" Stint {int(st)} | {comp:7s} | {len(grp):2d} vong "
            f"(sach: {len(c):2d}) | vong "
            f"{int(grp.LapNumber.min())}-{int(grp.LapNumber.max())}"
        )

    out = laps[laps["clean"]][
        ["LapNumber", "Stint", "Compound", "TyreLife", "LapTimeS"]
    ]

    fname = f"{DRIVER}_{YEAR}_{gp}_clean.csv"
    out.to_csv(fname, index=False)
    print(f" -> da xuat {len(out)} vong sach: {fname}")


def main():
    for gp in GPS:
        extract_one(gp)


if __name__ == "__main__":
    main()
```

7.2 A6(b). Estimating the Parameters from Real Data

Each clean lap has the form

$$(r, \ell, c, k, y)$$

where r is the race, ℓ is the global lap, c is the compound, k is the tire age within the stint, and y is the observed lap time.

Observation model after separating fuel and track

If we fit the original model directly to y, we are assuming that the lap time depends only on the tire age k. This is not true for a real race. A lap at the start of the race is usually slower because the car is heavy with fuel; a lap at the end of the race may be faster because the track has more rubber laid down.

Therefore we use the observation model:

$$y_i = B^{\text{ref}}_{r_i,c_i} + d_{c_i}(k_i - 1) + F_{r_i}(\ell_i) + H_{r_i}(\ell_i) + \varepsilon_i$$

Table 6: Meaning of each component in the observation model.

Symbol	Meaning	Unit
$B^{\text{ref}}_{r,c}$	Base lap time of a new tire of compound c at track r, at the reference point: the car nearly out of fuel and the track already evolved.	seconds
d_c	Tire degradation rate before the cliff, after separating out fuel and track.	seconds/lap
$F_r(\ell)$	Penalty due to the remaining fuel at global lap ℓ. Large at the start of the race, near 0 at the end.	seconds
$H_r(\ell)$	Penalty due to the track not yet being fully rubbered in. Larger at the start of the race, then decreasing.	seconds
ε_i	Remaining noise: minor traffic on track, measurement error, pace management, wind and temperature.	seconds

We write $B_{r,c}$ and not just B_c because Bahrain, Spain and Hungary have different lap lengths, so the base lap times differ by tens of seconds. Therefore B_S at Bahrain cannot be pooled directly with B_S at Spain. In multi-track data, the correct base pace must be $B_{r,c}$, or the data must first be normalized by track. In this report we estimate B by stint and by track, while d is pooled by compound as a descriptive indicator.

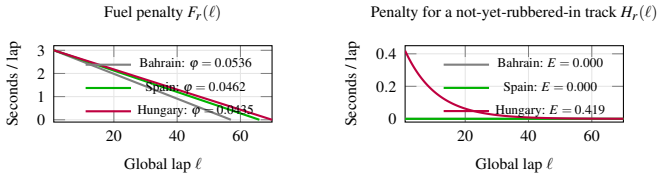


Figure 17: The two corrections subtracted from the observed lap time. Left: the fuel penalty decreases almost linearly. Right: the penalty due to the not-yet-evolved track decreases exponentially. In the main table, we use the track correction only when E_r can be quantified from the data, to avoid assigning arbitrary numbers.

How to Determine E_r Without Arbitrary Guessing

If a race has two stints on the same compound, with the same driver and on the same track, then after subtracting fuel and track evolution, the two base intercepts must be close to each other. For two stints on the same compound starting at ℓ_a, ℓ_b:

$$\widehat{B}^{\text{raw}}_a - F_r(\ell_a) - E_r e^{-(\ell_a-1)/\lambda} = \widehat{B}^{\text{raw}}_b - F_r(\ell_b) - E_r e^{-(\ell_b-1)/\lambda}.$$

It follows that

$$\widehat{E}_r = \frac{\left(\widehat{B}^{\text{raw}}_b - F_b\right) - \left(\widehat{B}^{\text{raw}}_a - F_a\right)}{e^{-(\ell_b-1)/\lambda} - e^{-(\ell_a-1)/\lambda}}$$

where:

$$F_a = F_r(\ell_a), \qquad F_b = F_r(\ell_b), \qquad \lambda = 10.$$

7.3 A6(c). Evaluation of the Current Model and the Extended Baseline–Luck–Error Model

Part A6(c) extends the calibration model above by splitting the actual race time into three parts: the baseline part that can be computed with the tire–pit model, the deviation due to the race context, and the remaining error. Denote by T_{base} the time forecast by the baseline model after accounting for tires, pit stops, fuel, track evolution and the first lap. The decomposition model is then

$$T_{\text{actual}} = T_{\text{base}} + T_{\text{luck}} + \varepsilon, \qquad T_{\text{luck}} = T_{\text{skill}} + T_{\text{pure}}. \tag{16}$$

In forecasting, ε is not added directly because it is the error remaining after the model has made its forecast. Therefore the forecasting formula used in the experiments is

$$T_{\text{pred}} = T_{\text{base}} + \widehat{r}, \qquad \varepsilon = T_{\text{actual}} - T_{\text{pred}}. \tag{17}$$

Here $\widehat{r}$ is the residual part learned from the data. This formulation keeps the mathematics of A1–A4 as the physical core, while allowing the real data to explain deviations due to starting position, traffic, yellow flags, Safety Car, accidents, rival retirements, weather and track characteristics.

Role of the three components

T_{base} answers the question "if the race proceeds normally according to tires, pit stops, fuel and track, how long does it take?"; T_{luck} measures the part that is faster or slower due to the real context of the race; ε is the unexplained part, used to test the model rather than to hide the error.

7.3.1 Extended Baseline and Exponential Cliff

The baseline model is kept from A1–A4 but adds fuel, track evolution and the first lap:

$$T_{\text{base}} = \sum_{\text{laps}} t_c(k) + N_{\text{pit}} P + T_{\text{fuel/track}} + T_{\text{first lap}}. \tag{18}$$

The component $T_{\text{fuel/track}}$ combines two opposing trends: the car becomes lighter as it burns fuel, while the track usually becomes faster as more rubber is laid down. The component $T_{\text{first lap}}$ separates out the first lap, because this lap is strongly affected by the start, jostling and traffic.

What is new in A6(c) is allowing the part after the cliff to increase exponentially. Specifically,

$$t_c(k) = \begin{cases} B_c + d_c(k-1), & k \le L_c, \\ B_c + d_c(L_c - 1) + \dfrac{D_c}{\lambda_c}\left(e^{\lambda_c(k-L_c)} - 1\right), & k > L_c, \end{cases} \tag{19}$$

with $\lambda_c > 0$ the curvature after the cliff. As $\lambda_c \to 0$,

$$\frac{e^{\lambda_c(k-L_c)} - 1}{\lambda_c} \approx k - L_c,$$

so (19) returns to the old linear model after the cliff. Therefore this is a generalization, not a contradictory replacement of the original model.

For a stint of length n laps, if $n \le L_c$ then

$$S_c(n) = nB_c + \frac{d_c}{2} n(n-1).$$

If $n > L_c$, let $h = n - L_c$; we have

$$\begin{aligned} S_c(n) = {} & L_c B_c + \frac{d_c}{2} L_c(L_c - 1) + h\big(B_c + d_c(L_c - 1)\big) \\ & + \frac{D_c}{\lambda_c}\left[\frac{e^{\lambda_c}\left(e^{\lambda_c h} - 1\right)}{e^{\lambda_c} - 1} - h\right]. \end{aligned} \tag{20}$$

This formula is useful when the real data show that a stint is extended deep beyond the cliff.

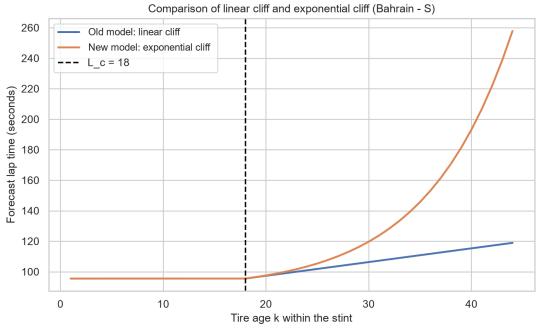


Figure 18: Comparison of the linear cliff and the exponential cliff. Before L_c, the two models are close; after L_c, the exponential model allows the lap time to increase faster when the tire falls out of its operating window.

7.3.2 Splitting Luck into Skill-Based Luck and Pure Luck

The residual of each observation i is defined by

$$r_i = T_{\text{actual},i} - T_{\text{base},i}.$$

This part is split into two groups for easier interpretation:

$$r_i \approx T_{\text{skill},i} + T_{\text{pure},i}.$$

Skill-based luck includes the factors related to the driver, the car or the team's decisions whose final effect depends on the race context: starting position, starting and overtaking ability, tire management, clean-lap consistency, car speed in clean air, pit-stop quality, pit timing, undercut/overcut and the appropriateness of the compound choice. *Pure luck* includes events that are largely exogenous: collisions, time lost on the first lap, rival

Table 7: Representative variable groups in the luck model. The expected sign is understood after the variables have been standardized.

Group	Representative variables and meaning
Skill-based luck	`starting_position`, `driver_start_gain`, `driver_overtake_gain`, `driver_tire_management`, `driver_consistency`, `car_pace_rank`, `pit_stop_quality`, `pit_timing_quality`, `strategy_compound_score`.
Pure luck	`incident_involved`, `incident_time_loss_estimate`, `first_lap_time_loss`, `rival_dnf_gain`, `track_overtaking_difficulty`, `safety_car_laps`, `yellow_flag_sector_loss`, `weather_change_indicator`.

DNF/DSQ, the overtaking difficulty of the track, Safety Car/VSC, yellow flags, being lapped, weather changes and mechanical failures.

The sign convention must be stated clearly: the variables are usually standardized so that a larger value corresponds to an increase in time. If a variable measures a benefit, such as `rival_dnf_gain` or `driver_overtake_gain`, its expected contribution must be negative or its sign must be flipped before it enters the model.

7.3.3 An Interpretable Residual Learning Model

A6(c) does not use a dense black-box network to forecast T_{actual} directly. Instead, the model only learns the residual r_i with a Neural Additive Model–type architecture:

$$\widehat{r}_i = \beta_0 + \alpha_s \sum_{j=1}^{p_s} w_{s,j} f_{s,j}(z_{s,i,j}) + \alpha_p \sum_{r=1}^{p_p} w_{p,r} f_{p,r}(z_{p,i,r}) + g_{\text{interaction}}(z_i), \tag{21}$$

where $z_{i,j} = (x_{i,j} - \mu_j)/\sigma_j$. Each f_j is a small single-variable network, so the contribution of each variable can still be read off; $g_{\text{interaction}}$ is only used to capture important interactions, for example

$$\texttt{start_pos} \times \texttt{overtaking_difficulty},$$
$$\texttt{safety_car_timing} \times \texttt{pit_window}.$$

From (17) and (21), the final forecasting formula is

$$T_{\text{pred},i} = T_{\text{base},i} + \widehat{r}_i. \tag{22}$$

7.3.4 Controlling Overfitting

F1 data are not large: a season usually has only about 20–24 races, each with about 20 drivers. Therefore a neural model is reasonable only if accompanied by strict controls:

1. **Do not relearn the baseline:** the model only learns $r_i = T_{\text{actual},i} - T_{\text{base},i}$, and does not forecast T_{actual} directly.
2. **Race-wise train-test split:** do not split randomly by driver-race, because drivers in the same race share weather, Safety Car/VSC and track conditions. Therefore

$$\text{Train races} \cap \text{Test races} = \varnothing.$$

3. **Outlier-robust loss function:**

$$\mathscr{L} = \frac{1}{n}\sum_{i=1}^{n} \rho_\delta(T_{\text{actual},i} - T_{\text{pred},i}) + \lambda_1 \sum_j |W_j| + \lambda_2 \|\theta\|_2^2, \tag{23}$$

where ρ_δ is the Huber loss, more robust than MSE in the presence of accidents, DNFs or abnormal laps.

4. **Early stopping and model comparison:** the neural additive luck model is accepted only if it improves MAE/RMSE on the race-wise test set compared with baseline-only and baseline + linear regression.

When interpreting, one should not look only at W_j, since the weights can be close to each other after standardization. Importance is more reliable for the question: if this variable is perturbed or removed, how much does the MAE increase? Therefore the weight table is only secondary evidence; the main conclusion must be based on the test error and the residual.

Table 8: Some variables with high importance in the A6(c) trial run. These values serve to illustrate how to read the model, not as universal physical constants.

Variable	Group	W_j	Importance
`rival_dnf_count_ahead`	pure	0.0991	0.0552
`qualifying_gap_to_teammate`	skill	0.1018	0.0438
`rival_dnf_position_value`	pure	0.1021	0.0438
`yellow_flag_sector_loss`	pure	0.1016	0.0412
`normalized_clean_race_pace`	skill	0.1027	0.0393
`safety_car_timing_benefit`	pure	0.1019	0.0360
`driver_overtake_gain`	skill	0.1041	0.0334
`traffic_management_score`	skill	0.1026	0.0315

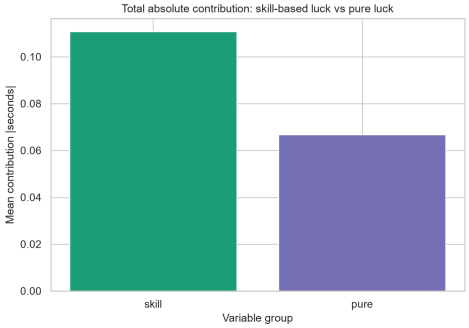


Figure 19: Illustration of how to compare the mean absolute total contribution of the skill-based luck group and the pure luck group. The conclusion about which group is larger depends on the set of races chosen, so the figure is meaningful only after checking on a race-wise test set.

7.3.5 Evaluation of the Old Model and the New Model

The additional part A6(c) requires a fair evaluation of the current model before proposing a new model. Therefore we treat the old model as the baseline-only model:

$$\widehat{T}_i^{(0)} = T_{\text{base},i}, \qquad e_i^{(0)} = T_{\text{actual},i} - \widehat{T}_i^{(0)}.$$

The reported metrics include

$$\text{MAE}_0 = \frac{1}{n}\sum_i |e_i^{(0)}|, \quad \text{RMSE}_0 = \sqrt{\frac{1}{n}\sum_i (e_i^{(0)})^2},$$
$$\bar{e}_0 = \frac{1}{n}\sum_i e_i^{(0)}, \qquad R_0^2 \text{ is reported alongside.}$$

If the residual $e_i^{(0)}$ still has structure with respect to the race context, the old model is not sufficient for race-level forecasting. A direct check is to regress the residual:

$$e_i^{(0)} = \gamma_0 + \gamma_1 \texttt{starting_position}_i + \gamma_2 \texttt{SC/VSC}_i + \gamma_3 \texttt{incident}_i + \cdots + u_i.$$

If the coefficients above have reasonable signs and reduce the error, we should not force every deviation into B_c, d_c, L_c, D_c; instead we should keep the physical baseline and learn the residual part separately.

The new model uses the same baseline but additionally forecasts the residual:

$$\widehat{T}_i^{(1)} = T_{\text{base},i} + \widehat{r}_i, \qquad e_i^{(1)} = T_{\text{actual},i} - \widehat{T}_i^{(1)}.$$

The improvement is read by

$$\Delta\text{MAE} = \text{MAE}_0 - \text{MAE}_1, \quad \Delta\text{RMSE} = \text{RMSE}_0 - \text{RMSE}_1,$$
$$\Delta R^2 = R_1^2 - R_0^2.$$

Since $\widehat{r}_i$ is learned with an additive model, each variable still has its own contribution curve; this is why the report prefers the Neural Additive Model over a dense network that is hard to interpret.

7.3.6 Sensitivity of the Baseline and the Luck Variables

The local sensitivity of the physical part shows which parameter errors are the most dangerous. For a stint that has not passed the cliff,

$$S_c(n) = nB_c + \frac{d_c}{2} n(n-1),$$

so

$$\frac{\partial S_c(n)}{\partial B_c} = n, \qquad \frac{\partial S_c(n)}{\partial d_c} = \frac{n(n-1)}{2}, \qquad \frac{\partial T}{\partial P} = N_{\text{pit}}.$$

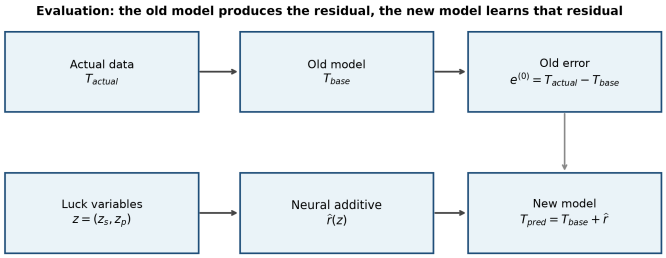


Figure 20: The A6(c) evaluation logic: the old model uses only T_{base}; the new model keeps the physical baseline and additionally learns the residual from the context variables.

If the stint has passed the exponential cliff, let $h = n - L_c > 0$; then

$$\frac{\partial S_c(n)}{\partial D_c} = \frac{1}{\lambda_c} \sum_{u=1}^{h} \left(e^{\lambda_c u} - 1 \right).$$

L_c alone is a discrete parameter, so the report uses the finite difference

$$\Delta_{L_c} T = T(L_c + 1) - T(L_c)$$

instead of a formal derivative. These formulas explain why a long stint close to the cliff is very sensitive to d_c, D_c and L_c, while a multi-pit strategy is directly sensitive to P.

With the residual model, if

$$\hat{r} = \beta_0 + \sum_j W_j f_j(z_j) + g_{\text{interaction}}(z),$$

then the sensitivity with respect to the original variable x_j is

$$\frac{\partial \hat{T}}{\partial x_j} = \frac{W_j}{\sigma_j} f_j'(z_j) + \frac{\partial g_{\text{interaction}}}{\partial x_j}.$$

In addition to the local derivative, the report uses permutation importance

$$I_j = \text{MAE}(X_{\pi(j)}) - \text{MAE}(X)$$

to measure the increase in error when the information of variable j is destroyed.

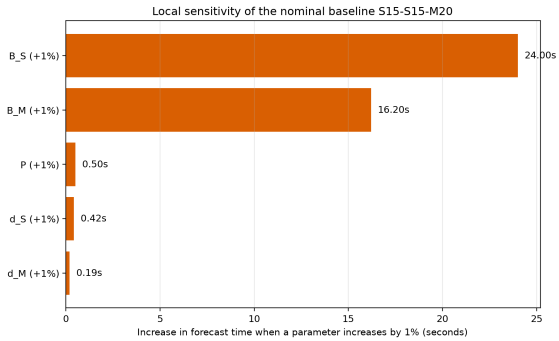


Figure 21: Local sensitivity of the baseline to B_c, d_c, D_c, L_c and P. The post-cliff parameters dominate strongly when the stint is extended through the cliff region.

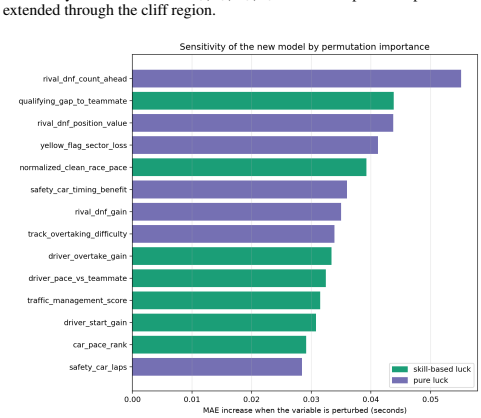


Figure 22: Permutation importance of the prominent context variables in the residual model. Similar weights are not enough to draw conclusions; the increase in MAE after permutation is the main measure.

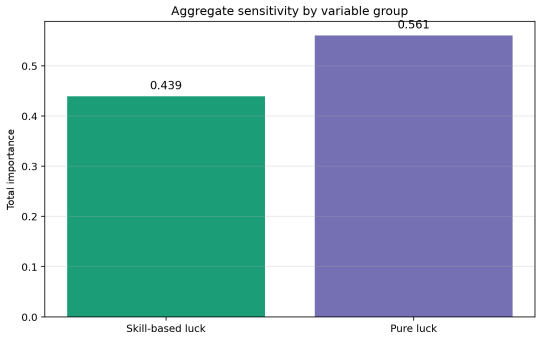


Figure 23: Comparison of contributions by variable group, skill-based luck and pure luck. This result must be read together with the race-wise test to avoid drawing conclusions from a small sample.

7.3.7 Robustness of the A6(c) Conclusions

Since race-level F1 data are small and many proxy variables are correlated, the A6(c) evaluation needs robustness checks instead of only reporting a single train-test split. The protocol used includes:

1. a race-wise train-test split, not a random split by individual driver;
2. leave-one-race-out to see whether the conclusions depend on a particular race;
3. race-wise bootstrap to estimate the standard deviation of MAE, RMSE and W_j;
4. changing the training seed and injecting small noise into the input variables;
5. ablation by group: removing skill-based luck, removing pure luck, or removing the interaction.

For each variable j, the weight stability is recorded by

$$\text{CV}(W_j) = \frac{\text{sd}(W_j)}{|\bar{W}_j| + 10^{-6}}.$$

A variable should be considered important only when it has both high permutation importance and a stable contribution sign across the data splits. If a variable stands out only in one race or one seed, the report classifies it as a weak signal.

7.3.8 Strengths and Limitations after the Additional Evaluation

The main strength of the new model is that it does not break the A1–A4 structure: the tire, pit, fuel and track parts remain in T_{base}, while the race-level factors are gathered into a testable residual. This separation helps evaluate the old model fairly: if the old model is wrong because of the Safety Car, an incident or the starting position, we see that error in the residual instead of arbitrarily modifying the tire parameters.

The limitations must also be stated clearly. F1 data are small; many variables such as overtaking ability, traffic quality or the benefit from rival DNFs are only proxies; the context variables are easily correlated; and D_c, λ_c are hard to estimate if teams usually pit before the tire falls out of its operating window. Therefore the A6(c) model is a framework for evaluating and improving forecasts, not a tool that replaces the entire real-time strategy system of an F1 team.

7.3.9 Plausibility Checks and Conclusion for A6(c)

The extended model must pass four basic checks. First, if every luck variable equals 0 and $\beta_0 \approx 0$, the model returns to the baseline. Second, as $\lambda_c \to 0$, the exponential cliff returns to the linear cliff, so it does not contradict the old model. Third, the signs of the variables must be reasonable: time loss due to an incident, a slow pit stop or a chaotic first lap increases the time; the benefit from overtaking or rival DNFs must decrease the time if it is measured as a benefit. Fourth, the residual must be concentrated around 0; outliers need to be explained by DNFs, being lapped, mechanical failures or missing data.

Conclusion of A6(c)

The proposed model is a verification framework:

$$T_{\text{actual}} = T_{\text{base}} + T_{\text{skill}} + T_{\text{pure}} + \varepsilon,$$

while the forecasting formula is

$$T_{\text{pred}} = T_{\text{base}} + \widehat{T}_{\text{skill}} + \widehat{T}_{\text{pure}} + g_{\text{interaction}}(z).$$

The strength is that it keeps the mathematical foundation of A1–A4, while adding the real race context by clearly separating skill-based luck and pure luck. The limitation is that many variables are only proxies; incident data may be missing; and D_c, λ_c are hard to estimate if teams usually pit before the tire reaches the cliff.

References

Note: the part in square brackets indicates the sections of the report that are mainly supported by that source.

[1] Pirelli. *Formula 1 tyres / Slick tyres and compound range*. Official Pirelli Motorsport website. Accessed 26/06/2026. `https://www.pirelli.com/tyres/en-ww/motorsport/f1/tyres`. [A1, A2, A4: the Soft–Medium–Hard compound system, the grip–durability–degradation trade-off.]

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8 Appendix

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